

Delegation as Incentive for Public Good Provision: Evidence from an Online Community *

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Abstract

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1 Introduction

Nowadays, productivity relies heavily on information crowdsourced from internet users and aggregated by knowledge platforms (e.g. Wikipedia and Stack Overflow) or large language models (e.g. OpenAI’s GPT). Software programmers rely on online sample code, journalists write pieces based on social media content, and restaurants rely on online reviews, to mention a few. While user-generated content is a valuable public good, its production is not remunerated nor contractible. How can platforms that host such content incentivise high-quality contributions? The literature has identified several non-monetary drivers of effort.¹ Platforms may exploit those that depend on their design as crucial devices to incentivise participation. Typical organisational tools are non-monetary rewards, like awards, that leverage users’ preferences for recognition and status (Gallus and Frey (2016)). Nevertheless, awards may incentivise effort before they are assigned, but not necessarily after. The theoretical literature in personnel economics has identified another non-monetary channel, that is, the delegation of decision rights.² In other words, platforms may incentivise participation through the strategic allocation of autonomy over decision-making (Gibbons, Matouschek, and Roberts (2013), Gambardella, Panico, and Valentini (2015)). Users may value autonomy when contributing because of either private benefits or because it gives them more control over outcomes. If that is the case, the platform may prefer to provide autonomy unconditionally to incentivise contributions (Aghion and Tirole (1997)).³ However, organizational designs where the allocation of autonomy is conditional on achieving performance targets may allow the platform to 1) focus users on specific tasks rather than others (Holmstrom and Milgrom (1991)), 2) create virtual *promotions* and the associated *career* incentives (Holmström (1999)), and 3) screen the best users across time (Gershkov and Perry (2012)).

In this paper, I investigate whether the delegation of decision rights leads to an increase in quantity and quality of online contributions. I identify whether and to what extent users are interested in obtaining more autonomy over tasks and study its role in contribution patterns. Using data from Stack Exchange, a question and answer (Q&A) website, I show that people value such autonomy. However, differ-

¹Drivers of effort include intrinsic utility and firm recognition (Roberts, Hann, and Slaughter (2006), Nov (2007), Ma and Agarwal (2007), Jeppesen and Frederiksen (2006)), the community size (Zhang and Zhu (2011)), reference points on others’ behavior (Chen, Harper, Konstan, and Li (2010)), within-community reputation (Chen, Ho, and Kim (2010)), peer recognition (Jin, Li, Zhong, and Zhai (2015), Chen, Wei, and Zhu (2017)), awards (Gallus and Frey (2016)), sequential targets (Goes, Guo, and Lin (2016)), and the signaling of skills (Belenzon and Schankerman (2015), Xu, Nian, and Cabral (2020)).

²In this paper, I interchangeably use the terms “decision rights”, “control rights”, and “autonomy”.

³This implication requires sufficiently close objectives of the platform and the users.

ent types of users value having such autonomy differently. Through counterfactual exercises, I explore organisational implications and find that delegation increases the number and quality of contributions, suggesting that more autonomy to users boosts content production. However, the delegation of autonomy conditional on achieving performance targets allows the platform to change the relative contribution of the different user types. The online community splits into a smaller, more committed group of users who are more likely to sustain contributions over time, and a larger group that is initially more engaged but more likely to become inactive (*enthusiasts*). Conditioning delegation on performance, the platform increases the relative contribution of *enthusiasts*, as they achieve the performance target faster.

What does it mean to allocate autonomy in digital platforms? Knowledge platforms need to incentivise effort on two main tasks: content production and the moderation of existing content (Gillespie (2018)). While the production of content is generally unrestricted, platforms differ in how much autonomy they grant users for moderation. Facebook does not allow users to modify content and hires professional moderators. Users are only allowed to flag content that they believe violates Facebook’s rules. In contrast, Wikipedia allows every internet user to modify existing articles. Finally, Stack Exchange provides autonomy in editing content conditional on achieving given performance targets. What trade-offs affect this decision?

To study this question, I use data from Stack Exchange. Stack Exchange is a family of websites where registered users ask questions and provide answers on different topics. Users can collect *reputation points*, mostly through community upvotes on their contributed answers. The moderation of the website relies on moderators elected within the community and community members’ edits. Users have control over the implementation of their edits if they have a number of reputation points larger than a threshold $\bar{R}$. If users have fewer reputation points than that, their edits are *suggested*. To be implemented, they require approval from either the owner of the edited content or from users with more than the threshold number of points.

The data I use include the contribution histories (e.g. answers and edits) of participants in the English Language Learners website.⁴ This website is part of the Stack Exchange family and focuses on questions concerning the use of the English language.

The analysis proceeds in two steps. First, I study the impact of variation in autonomy on users’ propensity and intensity in editing. On the one hand, I estimate a regression discontinuity with staggered treatment around the reputation threshold $\bar{R}$ to show that both the number of edits and the likelihood of editing increase once users obtain autonomy on editing. On the other hand, I exploit a

⁴<https://ell.stackexchange.com/>

variation of the reputation threshold with retroactive effect, which led to some users losing autonomy. Consistent with the previous result, users who lose control are less likely to edit and produce fewer edits.

Second, I use a dynamic discrete choice model (à la Rust (1987), but with continuous state space) to quantify users' marginal utility from participation and simulate counterfactuals. At each week of participation, users decide their contribution in terms of the number of answers, the quality of answers, and the number of edits. The utility function allows preferences to depend on the degree of autonomy by interacting the drivers of the utility with a dummy equal to 1 if the user gained autonomy. The identification relies on revealed preference and an inter-temporal tradeoff generated by time discounting. Indeed, users trade off contributions in the current week versus the following week, and prefer contributing in the current week if the discounted expected returns from contributing (e.g. reputation points and autonomy) outweigh the costs. Methodologically, I estimate the utility parameters using a conditional-choice-probability (CCP) based estimator with *finite dependence*, a tool that allows substantial computational gains as it does not require a full solution of the model (Arcidiacono and Miller (2011)).

The results show a positive marginal utility of autonomy and a significant increase in willingness to participate in editing once endowed with control over the action. Interestingly, users' utility of answering increases as well when users gain control over editing, suggesting positive spillovers of autonomy across tasks. The estimation identifies two unobserved types, to whom I will refer as *committed* and *enthusiasts*. *Committed* users are relatively few (about 20% of the sample), they are more likely to sustain contributions across time, and are possibly more extrinsically motivated, as they tend to disclose more information in their profile page. *Enthusiasts* instead form a larger group that is more active in the short run, but also more likely to become inactive with time. They differ on two key dimensions. First, *committed* users have higher initial costs in editing, but which are marginally decreasing faster. Second *committed* users value autonomy more when editing.

With estimates from the model, I simulate counterfactual contribution histories under delegation designs that differ on the threshold of reputation points $\bar{R}$. In particular, I consider the case with a performance threshold equal to zero (full delegation), 500, 1000, and 2000.

The results show that the platform would maximise quality and quantity of contributions with full delegation and without conditioning autonomy on performance. This is because both types of users make more effort if they have autonomy. However, over 150 weeks of participation, a positive performance threshold increases the relative contribution of *enthusiasts* with respect to *committed* users. Indeed, *enthusiasts* reach the threshold faster, meaning that, with a positive rep-

utation threshold, they have autonomy for a longer share of their contribution history. *Committed* users take more time to accumulate enough points to gain autonomy. However, they are less likely to become inactive, sustaining contribution levels much longer. With more demanding thresholds, this effect partly offset the previous one as, in the long run, *enthusiasts* are likely to become inactive before gaining autonomy.

Finally, I do not find that the quality of edits is significantly different across user types, suggesting that the dynamic selection across time does not necessarily help in screening low-ability types.

This paper contributes to the literature in several ways. First, I show direct evidence of non-monetary preferences for control and identify in real data the value of autonomy. This result confirms experimental and survey-based evidence showing that individuals value control rights and power (Fehr, Holger, and Wilkening (2013); Bartling, Fehr, and Herz (2014); Owens, Grossman, and Fackler (2014); Pikulina and Tergiman (2020); Meagher and Wait (2021)).

Second, the paper contributes to the empirical literature on the effects of providing autonomy on workers' performance. Research shows that in India, with more autonomy, procurement officers became more efficient (Bandiera, Best, Khan, and Prat (2021)) and managers increased firms' value added and markups (Kala (2024)). In Brazil, greater autonomy for teachers increased student learning and progression (Piza, Zwager, Ruzzante, Dantas, and Loureiro (2024)), while in Argentina, loan officers exerted more effort in production and the use of soft information (Liberti (2018)).⁵

Third, the paper contributes to the literature that studies user behaviour in online communities, in particular in Stack Exchange. Aaltonen and Wattal (2025) study how the type of rejection notices affects user retention; Xu et al. (2020) and Forderer and Burtch (2024) investigate career concerns, Bregolin (2025) studies how the use of a foreign language affects communication quality, and Burtch, Lee, and Chen (2024) study how the popularisation of large language models is affecting

⁵A substantial empirical literature has studied patterns of delegation in firms. De Varo and Prasad (2015) and Hong, Kueng, and Yang (2019) study the complementarity between delegation of control and performance-based monetary incentives. They show that it depends on job complexity (De Varo and Prasad (2015)) and that is inversely related to the delegation of control from managerial to non-managerial roles (Hong et al. (2019)). McElheran (2014) shows that delegation is more likely if the firm's adaptation to local problems is more important than within-firm coordination. Katayama, Meagher, and Wait (2018) find evidence that firms jointly optimise communication protocols and delegation of control decisions, and that delegation is more likely when valuable human capital is dispersed across the firm. Bloom, Sadun, and Van Reenen (2010), Meagher and Wait (2014), and Liu, Meagher, and Wait (2022) show that market conditions affect firms' incentives to delegate by making local knowledge more salient and affecting employees' trust in managers, which is associated with higher levels of delegation (Bloom, Sadun, and Van Reenen (2012); Meagher and Wait (2020)).

user contributions. However, none of these papers uses data from the ELL website.

Finally, while the results of this paper are specific to the context of online communities, they may suggest implications for a broader set of environments, addressing puzzles that emerged in the literature on promotions. They can provide a plausible explanation for the use of promotions rather than bonuses, even if bonuses are more flexible incentives (Baker, Jensen, and Murphy (1988), Gibbons and Waldman (1999)). In addition, they contribute to the investigation of the fact that often firms commit to promote employees on the grounds of observable measures not correlated to the skills required for the delegated tasks (Peter principle, Fairburn and Malcomson (2001), Benson, Li, and Shue (2019)).

The paper proceeds as follows. Section 2 describes the setting and the rationale for delegation, and section 3 presents the data. I then present the results from the reduced-form analyses in section 4 and the structural model in section 5. Section 6 reports the counterfactual analysis. Finally, section 7 concludes.

2 Stack Exchange: “Self Managed” Platforms

Stack Exchange is a family of 172 websites where users can freely and voluntarily ask and answer questions on a topic specific to each website. Participation does not involve any monetary transactions. The most well-known site is *Stack Overflow*, which hosts questions and answers about programming languages. These websites belong to a commercial company, which, as of July 2020, has raised 153 million dollars in venture capital and was sold in June 2021 for US\$1.8 billion.⁶ To give a sense of the welfare produced to consumers, in 2025, Stack Exchange received 418.8 million monthly visits and 806.3 million monthly page views. Users created 3.1 million questions, which received 3.5 million answers.⁷

Participation in Stack Exchange is subject to an incentive system based on virtual rewards, *badges* and *privileges*. Badges are comparable to medals or firms’ bonuses and depend on the accomplishment of given performance targets. Privileges instead provide access to additional resources or actions, and, in general, to a more influential role in the community. Users achieve them sequentially, by accumulating reputation points. For instance, with 15 points, users achieve the possibility to upvote other users’ contributions, while if they reach 20000 points, they achieve close to full administrative control of the site.⁸ Users obtain reputa-

⁶<https://www.businesswire.com/news/home/20200728005330/en/Stack-Overflow-raises-85M-Series-funding-accelerate> and <https://www.prosus.com/news-insights/group-updates/2021/prosus-to-acquire-stack-overflow>

⁷<https://stackexchange.com/about>

⁸The platform also delegates the website management through elections. At certain times, community members can vote to elect *moderators* who, once elected, jump to the top of the

tion points in several ways, mostly from upvotes on their questions and answers, and from getting their answers *accepted* as the one solving the question. Users can also get a few points when they make suggested edits to other users' content, and the edits get approved.⁹

2.1 Delegation of Control over Editing in Stack Exchange

The sequence of privileges that users can achieve by accumulating reputation points is comparable to a managerial hierarchy in the community. It allows the platform to delegate control over decision-making to volunteer users based on a performance indicator (reputation points). One particular example, which is the main focus of this paper, is the delegation of autonomy on editing content. Editing is the action of modifying existing questions and answers to improve them. Users can always make edits. However, if users have not collected enough reputation points to unlock the editing privilege, their edits are not directly implemented. They are rather suggestions, and need to be approved by the author of the modified content or by other users who already have the editing privilege. Users with the editing privilege, instead, can directly implement the edits. In other words, the platform uses the editing privilege to delegate control over the implementation of the edits.

Delegation of such control is instrumental to the platform in two possible ways. On one side, it allows for saving internal resources. By allowing users to voluntarily review and approve suggested edits, the platform can employ fewer moderators and exploit users' local knowledge. Indeed, if users' expertise is complementary to that of platform managers, an efficient design involves shared moderation responsibilities with the consumers (Roels (2014)).¹⁰ On the other side, delegation creates participation incentives if users value autonomy.

The theoretical and experimental literature suggests that individuals may value autonomy either for its intrinsic value and private benefits, or because it gives them control over outcomes that would otherwise differ from their preferred ones.¹¹

privilege hierarchy even if they do not meet the reputation requirement. Elected *moderators* keep their role permanently.

⁹In the supplementary appendix 1, figure 2 lists the rules to gain points, while table 1 reports the list of privileges and the reputation points necessary to obtain each of them.

¹⁰Crowd-based platforms require substantial moderation resources. In 2020, Facebook employed about 15000 moderators who, however, were considered insufficient: Charlotte Jee, MIT Technology Review, June 2020. More recently, a cost-saving reduction in the number of moderators of the platform X (previously Twitter) raised numerous concerns (<https://www.theguardian.com/technology/2022/oct/28/twitter-takeover-fears-raised-over-disinformation-and-hate-speech>)

¹¹For example, Bartling et al. (2014), Fehr et al. (2013), and Pikulina and Tergiman (2020) find in experimental settings that individuals enjoy the intrinsic value of control. Beckmann and

Under either alternative, users are more willing to contribute if endowed with control. The platform can provide users with autonomy to incentivise their effort. Under this consideration, and unless the platform believes that users' preferences diverge from those of the platform, the theory would suggest the optimality of full delegation of control.

However, two features of online knowledge platforms may lead to alternative optimal designs. First, users contribute to two tasks: answering and editing. If the platform believes that one is more important than the other, it may optimally design stronger incentives on that task than the other (Holmstrom and Milgrom (1991); Dewatripont, Jewitt, and Tirole (2000)). For instance, in Q&A websites like Stack Exchange, answering contributions may be first order. From this perspective, full autonomy on answering and autonomy based on performance on editing create asymmetry of incentives and may lead users to focus more on answering than editing. This effect is expected to be even stronger if the performance indicators are based on successful answering.¹²

Second, there is no formal contract that ties users to participate across time, nor that could be rescinded if contributions are of low quality. Platforms need to retain users, while possibly learning users' abilities and selecting better types across time. In this context, the delegation of autonomy based on performance can be beneficial for two reasons. On the one hand, it creates *career* incentives (Holmström (1999)). Users may want to contribute on the platform to show they are worth getting autonomy, or to signal their skills outside of the community (Xu et al. (2020); Forderer and Burtch (2024)).¹³ On the other hand, it may create incentive for only high-ability users to stay active on the platform, as the reward of autonomy will be granted only conditional to a path of successful achievements feeding into the performance measure (Gershkov and Perry (2012)).¹⁴

Kräkel (2022) find evidence of higher task commitment if workers have autonomy. Meagher and Wait (2021) find evidence of preference for power, particularly for people from individualistic cultures. Aghion and Bolton (1992) assume that entrepreneurs receive private benefits from control. The theoretical literature that studies delegation with principal-agent models relies on the assumption that more autonomy allows the agent to affect the outcome in their preferred way and, as a consequence, incentivise their effort (Aghion and Tirole (1997); Mookherjee (2006); Bester and Krämer (2008); Hart (2017). Finally, Sturm and Antonakis (2015) underline how power affects the direct inclination to take an action.

¹²Asymmetric incentives are instead suboptimal if the platform cares about both tasks equally. Since performance on editing is harder to measure than for answering, incentives based on performance should be avoided at all (Holmstrom and Milgrom (1994))

¹³If the signalling outside of the platform is the major driver, it may still be preferable to delegate control unconditional on performance, as outcomes would be more informative of users' skills (Blanes i Vidal (2007))

¹⁴The literature has also addressed other types of nonmonetary incentives, that have similar features to the delegation of control rights. Auriol and Renault (2008) and Besley and Ghatak (2008) investigate status incentives, while the tournaments literature has studied promotion

Overall, the optimal delegation decision for the platform is not obvious. The provision of autonomy on both tasks should incentivise participation overall. However, the alternative design where users have autonomy on answering, but need to achieve performance targets to obtain autonomy on editing, may 1) induce users to focus more on answering rather than editing, 2) create additional *career* incentives before autonomy is awarded, and 3) select the best users in the long run. The intensity of these effects may then depend on how demanding the performance target is, i.e. how many reputation points the platform requires to delegate autonomy. The counterfactual analysis will vary such requirements to quantify the platform tradeoffs.

3 Data

In this paper, I use data from the Stack Exchange website called *English Language Learners* (ELL), which focuses on questions and answers related to the use of English. This specific website is particularly suitable for the analysis for two reasons. First, posts contain only text, not equations or scripts, as in more technical Q&A websites. This allows us to measure the quality of the answers with text measures. Second, in the middle of the sample period, the site changed the reputation thresholds to achieve privileges, creating an additional variation on users' control over editing, as some users lost the *editing* privilege. Figure 1 reports the number of users with the privilege over time and shows that, in February 2016, 94 users lost the privilege.¹⁵

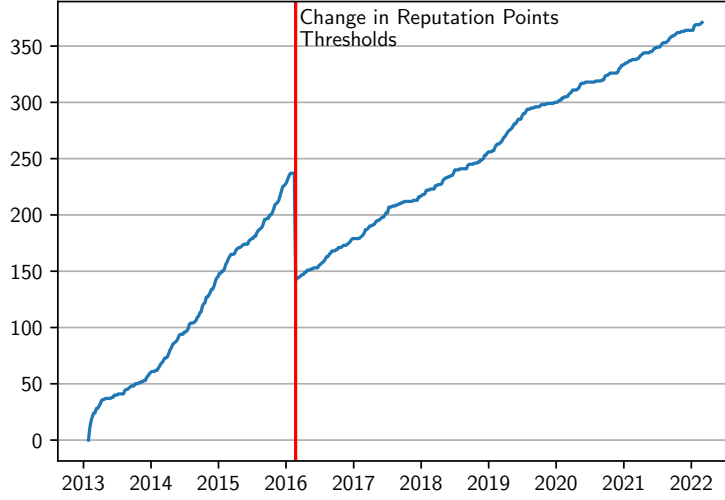
The data were retrieved on March 7th, 2022, and contain the complete set of user profile pages, contributions (e.g. answers and edits), and users' reputation histories. I constructed a panel of users' weekly participation in the website by including users who contributed at least a total of 5 answers and/or edits.¹⁶ Users' participation histories start with their first answer or edit and continue until the data retrieval date. The panel includes the weekly number of contributed answers and edits, and their average quality for 2625 users.

The measure of answer quality is an aggregated measure of several proxies for quality, namely the speed of response, measures of clarity, completeness, and informativeness assessed by a large language model (LLM), and text characteristics,

(Lazear and Rosen (1981)) and rank based incentives (Ehrenberg and Bognanno (1990)). These papers include rivalry between workers in obtaining status and promotions. In my work, instead, delegation does not depend on other workers' actions.

¹⁵Figure 1 in the supplementary appendix 1 reports the timing of the change within the sample period. Users knew that the change would happen, but did not know when exactly. Table 1 in the supplementary appendix 1 reports the number of reputation points required to obtain each privilege and how that changed after the new design.

¹⁶I include edits on questions' titles, questions' tags, and questions and answers' bodies.



In February 2016, an increase in the requirement of points to obtain this privilege induced the loss of the privilege for 94 users.

Figure 1: Number of users with the editing privilege

including length and number of links. In practice, the quality variable corresponds to the first component of the standardised and PCA-transformed set of proxy measures.¹⁷ The measure of edit quality is LLM-based and captures the number of issues fixed by the edit.¹⁸

The data are right-censored at the download date. Table 1 provides descriptive statistics of user activity. As it is standard in online communities, participation is skewed, with a relatively small group of users contributing a substantial part of the site content. Consequently, users are heterogeneous in terms of the number of reputation points achieved and whether they have reached the threshold.

¹⁷The first component explains 34% of the variance. Appendix A.1 provides details on the construction of the proxy variables and the aggregation method. Using different data, Harper, Raban, Rafaeli, and Konstan (2008) find that length and links are important predictors of answers' quality.

¹⁸Details on the measurement of edit quality are in appendix A.2.

	mean	std	min	median	max
Number of weeks in the sample	288.77	121.69	3.00	303.00	476.00
Amount of reputation points reached	1801.98	7402.57	6.00	414.00	179616.00
Number of answers	48.41	213.86	0.00	10.00	4850.00
Number of edits	25.84	228.71	0.00	0.00	6167.00
Average answer quality	6.11	0.92	2.78	6.17	8.35
Average edit quality (LLM)	1.16	0.85	-5.00	1.00	8.00
Reached editing privilege	0.16	0.37	0.00	0.00	1.00

Sample size: 2625 users. The variable *Reached editing privilege* takes the value 1 if the user reaches the threshold to achieve the *editing* privilege within the sample period, and zero otherwise. Statistics on answer and edit quality are conditional on a positive number of contributed answers and edits respectively.

Table 1: User-level descriptive statistics

4 Preference for Control

Reduced-form evidence shows that users are more willing to contribute edits after achieving the editing threshold. This is possible to see in figure 2, which reports estimates of the following regression:

$$Y_{it} = \alpha_i + \alpha_t + \beta_{r_{it}-\bar{R}} + \mathbf{W}'_{it}\rho + \varepsilon_{it} \quad (1)$$

where i indexes users, and t indexes weeks. Y is either the number of edits (left graph in figure 2) or a dummy equal to 1 if the user contributed any edits (right graph in figure 2). α_i identifies the user fixed effect, α_t the calendar week fixed effect, r_{it} the number of reputation points that user i has in week t , and $\bar{R}$ the number of reputation points required to obtain the editing privilege (i.e. 1000 points before February 2016 and 2000 points after). The parameters of interest are $\{\beta_{r-\bar{R}}\}_{\forall r}$, which identify the fixed effects of being $r - \bar{R}$ points away from the threshold $\bar{R}$.¹⁹ Finally, I include control variables. One set of variables aims to control for other drivers of editing activity. It includes a dummy equal to 1 when the user is an elected moderator, a dummy equal to 1 when the user is a candidate in a moderator election, and dummies equal to one in the week the user has achieved editing-related badges.²⁰ A second set of variables aims to control for the user’s time availability and includes the number of answers and comments produced.

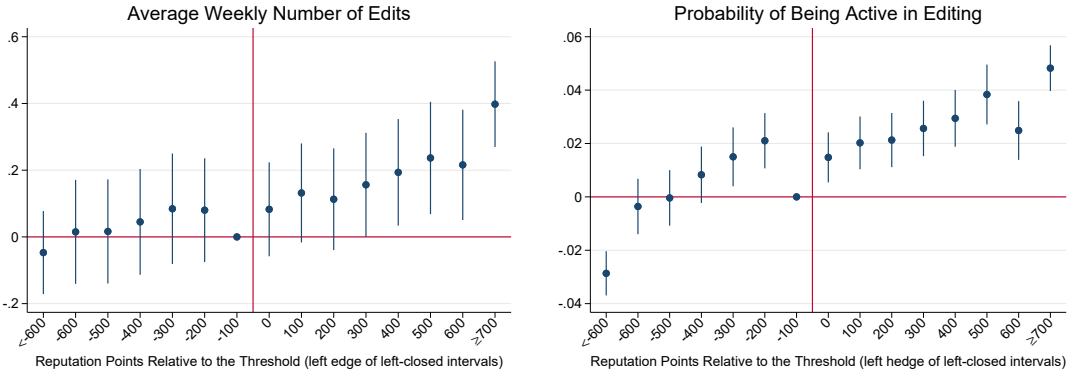
¹⁹In practice, I bin the reputation points missing to reach the thresholds in 100-points intervals, with the first and last bins defined such that $r - \bar{R} < -600$ and $r - \bar{R} \geq 700$ respectively.

²⁰Badges are sort of virtual medals. The editing-related badges are the *Copy Editor* and *Strunk & White* badges (<https://ell.stackexchange.com/help/badges>).

For the reduced form analysis only, I exclude users who did not reach the threshold to prevent selection effects.

Figure 2 reports estimates of the parameters $\{\beta_{r-\bar{R}}\}_{\forall r}$. They show that users are more likely to participate in editing content and edit more when they have the *editing* privilege. These results suggest that users prefer contributing edits if they have full control over their implementation.²¹ However, more autonomy does not imply higher quality. Indeed, a similar analysis that studies edit quality as an outcome variable does not find significant effects, as reported in figure 9 in appendix B.3.

A reduced-form analysis with standard tools cannot identify the effect of the threshold on answering behaviour as it cannot account for forward-looking behaviour, and the number of reputation points are endogenous to answering activity.²²



Reputation points fixed effects before and after achieving the editing privilege. Sample of users who reached the threshold.

Figure 2: Editing Contributions Relative to Achieving the Privilege

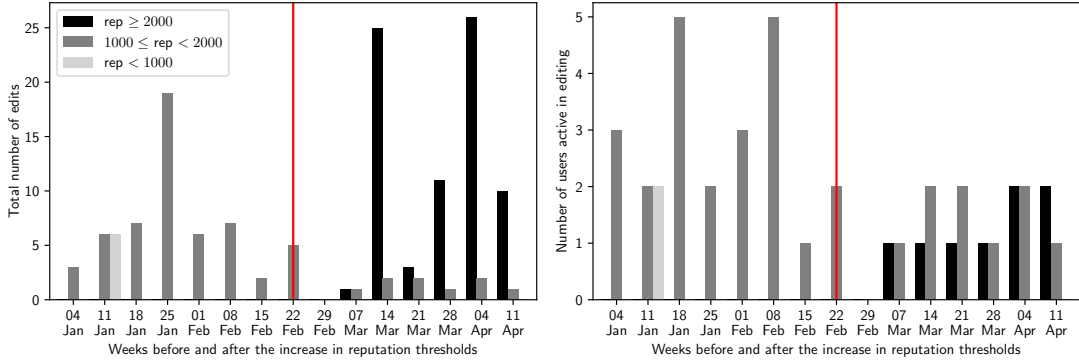
²¹Table 8 in appendix B.1 provides the complete set of estimates. It also provides estimates for a placebo regression where the outcome variable is the number of weekly comments written by the user. It provides evidence that contributions in a task unaffected by the editing privilege do not increase after users obtain autonomy in editing, suggesting that the increase in edits is attributable to the privilege.

²²Goes et al. (2016) address this problem relying on functional form assumptions and modifying the data to account for forward-looking behaviour. An estimation of model 1 with the number of contributed answers as an outcome variable suggests that users increase contributions and are more likely to participate in answering before they approach the threshold, as shown in appendix B.2.

4.1 Loss of Control

An alternative way to observe the role of control on the propensity to edit is to focus on users who lost control once the reputation threshold increased. Indeed, on February 24th, 2016, the editing threshold increased from 1000 to 2000. On that day, users with more than 1000 points but less than 2000 lost the editing privilege.

Figure 3 reports the editing activity of this specific group of users. It shows the total number of edits they made each week (left panel) and the number of users who made them (right panel). The colour identifies the number of reputation points the users had when they participated and contributed the edits, with the dark grey identifying users with control under the initial threshold but without under the new threshold. It is possible to notice that users stopped contributing edits right after they lost control. Editing activity then recovers, but it is driven by users who reached the new threshold (black bars).²³



Sample of 94 users who, on the week of February 22nd, 2016, had a number of reputation points ≥ 1000 and < 2000 . The left panel reports the number of edits they made each week (intensive margin), while the right panel the number of them who made the edits (extensive margin). Light grey bars are contributions of users with less than 1000 points, dark grey of users with at least 1000 points, but less than 2000, and black users with at least 2000 points. The vertical line identifies the week when the change in reputation thresholds took place.

Figure 3: Editing Contributions Around the Change of Reputation Threshold

²³In this context, a regression discontinuity design would not measure the parameter of interest, as users may reach the new threshold soon after the change in reputation thresholds. In addition, any more formal regression analysis would suffer from power issues due to the small sample size of the affected users (94 users).

5 Dynamic Discrete Choice Model

The reduced-form evidence has multiple limitations. First, it does not allow us to compare the incentive effect of allocating control relative to other types of motives. Second, it does not test for or quantify dynamic responses to the delegation of control. Third, it does not allow us to simulate counterfactual behaviour. Finally, it does not allow for investigating selection at the threshold.

To overcome these limitations, I develop a dynamic discrete choice model that accounts for forward-looking behaviour and unobserved heterogeneity on user types. Dynamic discrete choice models estimate preference parameters based on the concept of *revealed preferences*, that is, the assumption that choices are the outcome of (random) utility maximisation and, as such, provide information on users' preferences. In the context of participation in online communities, users choose their efforts to contribute to the platform. Their choice depends on the cost of effort, net of the choice's intrinsic utilities, and expected future benefits. Benefits could be, for instance, a certain number of reputation points or the achievement of privileges.²⁴

5.1 Users' Participation Choices

Users join the platform and start contributing on random dates. Their participation history starts when they first contribute an answer, a question, or an edit. Once on the platform, users cannot exit, but can decide not to contribute. For the sake of exposition, let T identify their last week of participation, which could be interpreted as their death.²⁵

In each week of participation $t \in \{t_0, \dots, T\}$, users decide their effort levels for two tasks, answering and editing. Effort is defined as a combination of the quantity and quality of answers and the quantity of edits. An action choice in week t is

²⁴Note that the application of dynamic discrete choice models to this context is conceptually similar to works that study dynamic investment decisions with discrete choice models. A typical application is human capital investment decisions. Examples of this literature are [Arcidiacono, Aucejo, Maurel, and Ransom \(2025\)](#) and [De Groote \(2024\)](#). Other examples of papers using this estimation strategy are [Yoganarasimhan \(2013\)](#) and [Khorunzhina \(2013\)](#), even though their context and exact application differ from this paper.

²⁵Exit from the site is observational equivalent to zero contributions in all following periods. For identification, users do not need to anticipate T . However, they need to believe that $T > t+3$ for any week t of participation. Note that, as I do not observe T , I assume $T > \mathcal{T}$, where $\mathcal{T}$ is the last week observed in the data, i.e. the week ending on March 6th, 2022

then a vector:

$$\alpha_t = \begin{bmatrix} NA_t \\ QA_t \\ NE_t \end{bmatrix} \in \mathcal{A}$$

where NA indicates the number of answers, QA denotes the average quality of answers, and NE indicates the number of edits. $\mathcal{A}$ represents the choice set, including all possible combinations of effort levels in the two tasks.²⁶

Each user i chooses an optimal sequence of choices to maximise the total sum of the discounted utility from all her periods of participation. Let $\alpha^* \equiv \{\alpha_t\}_{t < T}$ be such sequence of optimal choices. Then,

$$\alpha^* = \arg \max_{\alpha} \mathbb{E} \left[\sum_{t=t_0}^T \delta^{t-1} U_{\alpha t}(z_t) \right], \quad (2)$$

where δ is a discount factor and $U_{\alpha t}(z_t)$ is the flow utility that user i in states z_t receives in week t by choosing choice α .

Every week of participation proceeds as follows:

1. First, the user observes the values of the states realised at the end of the previous period, which include the total number of reputation points she has obtained, the number of questions available to answer, and her experience in terms of time spent on the website and the number of contributions. The number of reputation points implies how many privileges she has collected and whether she has already achieved the *editing* privilege.
2. Second, she forms beliefs over the value of the states that may be realised in the future, conditional on past choices and the possible new contribution(s) she could make. (Section 5.2 describes the assumptions on how users form beliefs.)
3. Third, she makes a contribution decision in editing and answering.
4. Finally, the flow payoff is realised and the states update to their new values.

The per-period flow utility from choosing choice α in week t , for a user of type θ , is defined as:

$$\begin{aligned} U_{\alpha t \theta} &= u_{\alpha t \theta}(z_t) + \varepsilon_{\alpha t}; \\ U_{\alpha t \theta} &= \beta_{0\theta} R_t + \beta_{1\theta} C_{\alpha t}^A + \beta_{2\theta} (C_{\alpha t}^A)^2 + \beta_{3\theta} C_{\alpha t}^E + \beta_{4\theta} (C_{\alpha t}^E)^2 + \beta_{5\theta} \text{cum} T_t \\ &\quad + \text{Control}_t (\beta_{6\theta} + \beta_{7\theta} R_t + \beta_{8\theta} C_{\alpha t}^A + \beta_{9\theta} (C_{\alpha t}^A)^2 + \beta_{10\theta} C_{\alpha t}^E + \beta_{11\theta} (C_{\alpha t}^E)^2) + \varepsilon_{\alpha t}, \end{aligned} \quad (3)$$

²⁶I discretise the effort levels to limit the computational burden. In practice, $NA \in \{0, 1, 7\}$, $QA \in \{0, 4.68, 6.61, 7.76\}$, and $NE \in \{0, 1, 9\}$, resulting in 21 possible combinations of efforts (with $QA > 0 \iff NA > 0$). More details are provided in appendix C.1.

where R_t are user's reputation points realised at the end of period $t - 1$, $cumT_t$ is the total number of privileges achieved, and $Control_t$ is a dummy equal to 1 if the user achieved the *editing* privilege. $cumT_t$ and $Control_t$ are deterministic functions of R_t and the privilege-thresholds system. The variables $C_{\alpha t}^A$ and $C_{\alpha t}^E$ are proxies for the amount of effort in answering and editing, respectively. They are defined as:

$$\begin{aligned} C_{\alpha t}^A &\equiv QA_{\alpha t} + NA_{\alpha t}^{scarcity_t}, \\ C_{\alpha t}^E &\equiv NE_{\alpha t}, \end{aligned}$$

where $NA^{scarcity}$ is the number of answers raised to a measure of the scarcity of questions to answer.²⁷ Finally, ε_{it} is an idiosyncratic choice-specific preference shock.

The parameter values are specific to an unobserved user type θ , which captures unobserved heterogeneity. The parameter β_0 captures the marginal utility of accumulating reputation points (R). A positive estimate would suggest that users either enjoy collecting points per se, as it would happen if they treat the site as a video game, or benefit from reputation points externalities (e.g. if points signal ability to employers). The parameters associated with C^A and C^E capture instead direct utility from making a certain contribution choice, including a cost of effort and the intrinsic benefit from the action (e.g. if the user is altruistic or enjoys participating per se). The variables R , C^A and C^E are interacted with $Control$ to allow for a change in user preferences once they obtain autonomy.

5.2 Beliefs

Users form beliefs and expectations over the evolution of the state space, given the contribution choices they make. In this section, I make assumptions on how users form such expectations.

5.2.1 Evolution of Reputation Points

Users gain and lose reputation points mostly through upvotes and downvotes on their content. Votes may arrive the same week the user publishes the answer or later. Suggested edits also provide reputation points if and when approved.²⁸

²⁷The variable *scarcity* is the inverse of the number of unanswered questions. It takes values in $[1, \infty]$ so that when there are many questions to answer, the cost of answering tends to be linear in the number of answers, while with fewer questions available, the cost becomes increasingly convex. Details on the construction of the *scarcity* variable are in appendix A.3.

²⁸Details on all possible sources of reputation points are in figure 2, supplementary appendix 1.

I assume users will expect to receive reputation points based on the quantity and quality of their content, and on their experience on the platform. I exclude other possible channels for the following reasons. First, voting is anonymous, suggesting that reputational effects that may bias the voting decisions are unlikely. Second, the site displays only the net sum of upvotes and downvotes. This limits bandwagon and anchoring effects, as users do not know precisely how peer contributors have voted. Third, peer effects and coordinated decisions are also implausible due to the large size of the community and the online nature of the interaction. Fourth, the content is objective and unopinionated, limiting voting for ideological reasons. Finally, while some more popular or successful users may attract more votes, I believe that the experience variables can partly capture such biases.²⁹

Consider the beliefs that the user forms in the first period of participation, t_0 . Each answer j that the user publishes in period t_0 receives, at the end of the period, a number of community edits and votes that follow a Poisson process:

$$\begin{aligned}\text{Received Edits}_{j,t_0+1} &\sim \mathcal{P}(\lambda_{E,j,t_0}), \\ \text{Up-votes}_{j,t_0+1} &\sim \mathcal{P}(\lambda_{U,j,t_0}), \\ \text{Down-votes}_{j,t_0+1} &\sim \mathcal{P}(\lambda_{D,j,t_0}).\end{aligned}$$

The expected values of these random variables are:

$$\lambda_{E,j,t_0} = \exp(\gamma_0 + \gamma_1 QA_{t_0} + \gamma_2 \text{Seniority}_{t_0} + \gamma_3 \text{Practice}_{t_0}), \quad (4)$$

$$\lambda_{U,j,t_0} = \exp(\gamma_4 + \gamma_5 QA_{t_0} + \gamma_6 \lambda_{E,j,t_0} + \gamma_7 \text{Seniority}_{t_0} + \gamma_8 \text{Practice}_{t_0}), \quad (5)$$

$$\lambda_{D,j,t_0} = \exp(\gamma_9 + \gamma_{10} QA_{t_0} + \gamma_{11} \lambda_{E,j,t_0} + \gamma_{12} \text{Seniority}_{t_0} + \gamma_{13} \text{Practice}_{t_0}), \quad (6)$$

where QA_{t_0} is the average quality of the user's answers published in period t_0 , and *Seniority* and *Practice* are measures of the user's experience. *Seniority* is the number of days the user has been participating on the website, and *Practice* is the cumulative number of answers she has published (both zero if $t = t_0$).

If the user, in her first period of participation, published NA_{t_0} answers, then she will expect to receive by the end of the period a number of up-votes and down-votes as follows:

$$\begin{aligned}\Lambda_{U,t_0} &= NA_{t_0} \lambda_{U,j,t_0}, \\ \Lambda_{D,t_0} &= NA_{t_0} \lambda_{D,j,t_0}.\end{aligned}$$

²⁹Users may also cast votes to achieve some badges, e.g, users achieve the *Civic Duty* badge if they vote 300 or more times. This may incentivise users to cast many votes inaccurately. However, badge incentives do not target specific users' content. For this reason, it is plausible that such "noisy" votes are unanticipated by contributors.

I model the number of approved edits, out of NE_{t_0} contributed suggested edits, as a binomial process, such that:

$$\text{ApprovedEdits}_{t_0+1} \sim \mathcal{B}(NE_{t_0}, \pi).$$

It follows that the expected number of reputation points (ρ) that the user expects to receive at the end of period t_0 is given by³⁰:

$$\mathbb{E}[\rho_{t_0+1}|\alpha_{t_0}] = 10\Lambda_{U,t_0}(NA_{t_0}, QA_{t_0}) - 2\Lambda_{D,t_0}(NA_{t_0}, QA_{t_0}) + 2\pi NE_{t_0}.$$

The answers produced in period t_0 may also induce the arrival of up-votes and down-votes in the following periods. I assume the process is deterministic and follows an exponential decay.³¹ Let Δt be the number of weeks passed from the publication week, such that if $t = t_0 + 1$, then $\Delta t = 1$. Then,

$$\begin{aligned}\lambda_{U,j,t_0+\Delta t} &= \lambda_{U,j,t_0} \exp\left(\frac{-\Delta t}{\tau_U}\right), \\ \lambda_{D,j,t_0+\Delta t} &= \lambda_{D,j,t_0} \exp\left(\frac{-\Delta t}{\tau_D}\right).\end{aligned}$$

$\lambda_{U,j,t_0+\Delta t}$ is the expected number of up-votes that the answer j , published in t_0 , receives in period $t_0 + \Delta t$, and similarly for down-votes. τ_U and τ_D are parameters.

If the user chooses positive effort in several periods, these processes aggregate. In general, the expected number of up-votes and down-votes arriving at the end of a given period t is, respectively,

$$\begin{aligned}\Lambda_{U,t} &= \Lambda_{U,t-1} \exp\left(\frac{-1}{\tau_U}\right) + NA_t \lambda_{U,j,t}(QA_t), \\ \Lambda_{D,t} &= \Lambda_{D,t-1} \exp\left(\frac{-1}{\tau_D}\right) + NA_t \lambda_{D,j,t}(QA_t),\end{aligned}$$

and the expected number of points arriving at the end of the period is

$$\mathbb{E}[\rho_{t+1}|\{\alpha_{\tilde{t}}\}_{\tilde{t} \leq t}] = 10\Lambda_{U,t} - 2\Lambda_{D,t} + 2\pi NE_t.$$

To conclude, let R_t be the cumulative number of points that the user observes to have at the beginning of week t . Then, by the end of the week, the user expects to have:

$$\mathbb{E}[R_{t+1}|R_t, \{\alpha_{\tilde{t}}\}_{\tilde{t} \leq t}] = R_t + \mathbb{E}[\rho_{t+1}|\{\alpha_{\tilde{t}}\}_{\tilde{t} \leq t}].$$

³⁰One up-vote gives 10 points, one down-vote removes 2 points, and the approval of a suggested edit gives 2 points.

³¹Alternative assumptions give similar results. Appendix C.2 shows how different functional forms fit the data.

5.2.2 Evolution of Questions' Availability

The availability of questions to answer evolves linearly: $avail_{it} = avail_{it-1} + \nu_1$.

5.3 Identification

The identification of the parameters affecting the beliefs on upvotes, downvotes, and edits arrival on own posts exploits cross-sectional variation at the post level, in the first week of publication. The panel dimension of post histories instead identifies the rates at which upvotes and downvotes arrive in the weeks following the publication. Indeed, the data include how many upvotes and downvotes arrive on a post each week.

The identification of the flow payoff parameters relies on the concept of *revealed preferences*, that is, the observation of user effort choices allows us to infer users' preferences. In practice, it follows identification results in the literature on dynamic discrete choice models, including, but not exclusively, [Rust \(1987\)](#), [Hotz and Miller \(1993\)](#), [Magnac and Thesmar \(2002\)](#), [Arcidiacono and Miller \(2011\)](#), and [Arcidiacono and Miller \(2020\)](#).

5.3.1 Identification of flow payoff parameters - theory

Let the conditional value function $\nu_{\alpha t}(z_t)$ be the user's value function of making choice α in week t , net of the preference shock $\varepsilon_{\alpha t}$. In addition, assume, as it is standard in the literature, that the preference shock follows a Type 1 Extreme Value distribution. Given the maximization problem stated in equation 2 and based on results from the literature ([Arcidiacono and Miller \(2019\)](#)), it follows that:

$$\begin{aligned} \nu_{\alpha t}(z_t) = & u_{\alpha t}(z_t) \\ & + \sum_{\tau=t+1}^{\mathcal{T}} \sum_{k \in \mathcal{A}} \sum_{z_\tau \in \mathcal{Z}} \delta^{\tau-t} (u_{k\tau}(z_\tau) - \ln(p_{k\tau}(z_\tau)) + \gamma) d_{k\tau}(z_\tau, d_{\alpha t} = 1) \kappa_\tau(z_\tau | z_t, d_{\alpha t} = 1) \\ & + \mathcal{V}_\alpha, \end{aligned} \tag{7}$$

where $\mathcal{T}$ is the last period in the sample, $p_{\alpha\tau}(z_\tau)$ is the probability of choosing α when in state z_τ (so called *conditional choice probability*, CCP hereafter), $d_{\alpha t}$ is a dummy equal to 1 if the user chooses α in week t , and $\kappa_\tau(z_\tau | z_t, d_{\alpha t} = 1)$ is the cumulative probability of being in state z_τ in week $\tau > t$ conditional on having

been in state z_t and having chosen α in week t :

$$\begin{aligned} \kappa_{\tau+1}(z_{\tau+1}|z_t, d_{\alpha t} = 1) \\ \equiv \begin{cases} f_{\alpha t}(z_{t+1}|z_t) & \text{for } \tau = t \\ \sum_{z_\tau \in \mathcal{Z}} \sum_{\mathbf{k} \in \mathcal{A}} d_{k\tau} f_{k\tau}(z_{\tau+1}|z_\tau) \kappa_\tau(z_\tau|z_t, d_{\alpha t} = 1) & \text{for } \tau = t+1, \dots, T. \end{cases} \end{aligned}$$

Finally, γ is Euler's constant and $\mathcal{V}_\alpha$ is the remaining utility that comes from participating after week $\mathcal{T}$ and until T . This latter part of the conditional value function is not identified as the sample ends at $\mathcal{T}$. However, identification is recovered using a property called *finite dependence*, as explained below.

The result in equation 7 show that for any choice α in week t , the future value term of the conditional value function can be expressed relative to any sequence of arbitrary choices. Consider two possible sequences of arbitrary choices that start in week t with choice α and α' respectively. As defined in Arcidiacono and Miller (2019), the pair of choices $\{\alpha, \alpha'\}$ exhibits r -period [finite] dependence if there exists sequences of decision[s] [...] from α and α' such that:

$$\kappa_{t+r+1}(z_{t+r+1}|z_t, d_{\alpha t} = 1) = \kappa_{t+r+1}(z_{t+r+1}|z_t, d_{\alpha' t} = 1)$$

In other words, *finite dependence* holds if a sequence of arbitrary contribution choices starting with α and one starting with α' would lead the user to given state values with the same probability. In that case, it is possible to write the differenced value functions of those choices as:

$$\begin{aligned} \nu_{\alpha t}(z_t) - \nu_{\alpha' t}(z_t) \\ = u_{\alpha t}(z_t) - u_{\alpha' t}(z_t) \\ + \sum_{\tau=t+1}^{t+r} \sum_{k \in \mathcal{A}} \sum_{z_\tau \in \mathcal{Z}} \delta^{\tau-t} (u_{k\tau}(z_\tau) - \ln(p_{k\tau}(z_\tau))) [d_{k\tau}(z_\tau, d_{\alpha t} = 1) \kappa_\tau(z_\tau|z_t, d_{\alpha t} = 1) \\ - d_{k\tau}(z_\tau, d_{\alpha' t} = 1) \kappa_\tau(z_\tau|z_t, d_{\alpha' t} = 1)] \end{aligned} \quad (8)$$

Note that, as long as $\mathcal{T} > t+r$, $\mathcal{V}_\alpha = \mathcal{V}_{\alpha'}$. The unidentified part of the future utility drops out of the difference.³² Given the assumptions on the flow utility function and transition probabilities, and sufficiently long data to be able to estimate the CCPs in a first step, all components of equation 8 are identified.

The exploitation of this result has two major advantages. First, it is not necessary to specify a terminal period, allowing for lighter assumptions on users' exit from the site. Second, the computational requirements are considerably lower than with alternative methods.³³

³²In the current paper, $r = 3$.

³³Some traditional alternative methods, like solving for the value functions via backward re-

5.3.2 Identification of flow payoff parameters - application

In the specific context of this paper, *finite dependence* holds because of the decreasing arrival of reputation points in the weeks following a contribution (see section 5.2). As a consequence, when users contribute, they expect it to produce reputation points only for a few weeks. As an example, let $\tilde{\alpha}$ be a possible contribution choice that leads, in expectation, to more reputation points in the following 3 weeks. Let also $\alpha^0 \equiv [0, 0, 0]$ identify the choice of no participation. Consider now the following two sequences of arbitrary choices: 1) $\{\tilde{\alpha}_{t_0}, \alpha_{t_0+1}^0, \alpha_{t_0+2}^0, \alpha_{t_0+3}^0, \alpha_{t_0+4}^0\}$; 2) $\{\alpha_{t_0}^0, \tilde{\alpha}_{t_0+1}, \alpha_{t_0+2}^0, \alpha_{t_0+3}^0, \alpha_{t_0+4}^0\}$. Following either sequence, the user would have, in expectation, the same amount of reputation points at $t_0 + 4$. Indeed, the second sequence just shifts the arrival of the reputation points by one week.³⁴ In this case, we would say that $\{\tilde{\alpha}, \alpha^0\}$ exhibits 3-period dependence. Given this example and the flow utility specification (eq. 3), equation 8 would rewrite as (omitting the squared terms of the flow utility for simplicity):

$$\begin{aligned}
& \nu_{\tilde{\alpha}t_0}(z_t) - \nu_{\alpha^0t_0}(z_t) \\
& \approx \beta_1(1 - \delta)C_{\tilde{\alpha}}^A + \beta_3(1 - \delta)C_{\tilde{\alpha}}^E + \beta_8Control_{t_0}(1 - \delta)C_{\tilde{\alpha}}^A + \beta_9Control_{t_0}(1 - \delta)C_{\tilde{\alpha}}^E \\
& + \beta_0 \sum_{\tau=t_0+1}^{t_0+3} \delta^{\tau-t_0} \mathbb{E}^{\tilde{\alpha}t_0}[R_{\tau} - R_{\tau-1}] + \beta_5 \sum_{\tau=t_0+1}^{t_0+3} \delta^{\tau-t_0} \mathbb{E}^{\tilde{\alpha}t_0}[cumT_{\tau} - cumT_{\tau-1}] \\
& + \beta_6 \sum_{\tau=t_0+1}^{t_0+3} \delta^{\tau-t_0} \mathbb{E}^{\tilde{\alpha}t_0}[Control_{\tau} - Control_{\tau-1}] \\
& + \beta_7 \sum_{\tau=t_0+1}^{t_0+3} \delta^{\tau-t_0} \mathbb{E}^{\tilde{\alpha}t_0}[Control_{\tau}R_{\tau} - Control_{\tau-1}R_{\tau-1}] \\
& - \delta (\ln(p_{\alpha^0}(\mathbb{E}^{\tilde{\alpha}t_0} R_{t_0+1})) - \ln(p_{\tilde{\alpha}}(\mathbb{E}^{\tilde{\alpha}t_0} R_{t_0}))) \\
& - \sum_{\tau=t_0+2}^{t_0+3} \delta^{\tau-t_0} (\ln(p_{\alpha^0}(\mathbb{E}^{\tilde{\alpha}t_0} R_{\tau})) - \ln(p_{\alpha^0}(\mathbb{E}^{\tilde{\alpha}t_0} R_{\tau-1})))
\end{aligned} \tag{9}$$

where $\mathbb{E}^{\tilde{\alpha}t_0}[z_{\tau}]$ captures the user's expected value of the state in week τ having contributed $\tilde{\alpha}$ in week t_0 and not participated after (i.e. having chosen the first

cursion, would be unfeasible in this setting, where the panel dimension extends up to 470 weeks. To my knowledge, this is the first paper that implements a dynamic discrete choice model with such a long time dimension.

³⁴In practice, this equivalence is only approximate because some parts of the utility flows are time dependent: the return of effort depends on the number of weeks spent on the website, the cost of answering depends on the number of available questions, and the CCPs are allowed to be time-dependent. However, the change of these time-dependent variables across two consecutive weeks is minimal. In this example, I abstract from these time dependences and remove the time indices on effort levels and CCPs.

arbitrary sequence of choices). The last two lines of the equation are offset terms that compensate for the chance that the arbitrary choices in the sequence are not optimal.

Based on this intuition, it is possible to construct similar differenced value functions for any pair of choices $\{\alpha, \alpha^0\}_{\alpha \in \mathcal{A}}$, where higher effort choices require longer sequences of no participation.

5.3.3 Identification of flow payoff parameters - identifying variation

Equation 9 clarifies the source of variation that identifies each flow utility parameter. Since the differenced conditional value functions (CVFs) capture the additional utility of making a contribution in a week rather than the next, identification relies on the existence of an inter-temporal trade-off. The identification of the direct net utility of contributions (i.e. C^A and C^E and their interaction with *Control*) requires imposing $\delta < 1$ and uses intuitive variation, with larger effort choices inducing larger values for those variables. The other variables are all functions of the total number of reputation points collected by the user (i.e. R). Since contribution choices impact them only in the future, their identification relies on variation in the future value terms. The variation comes from the total amount of points that users expect to receive, which depends on the levels of effort and experience. Since *cumT* and *Control* are discrete variables, in the differenced CVFs, there is identifying variation only if the choice allows the user to reach one of the reputation thresholds for which those variables vary in the periods of the finite dependence path. As a consequence, only observations with initial levels of reputation points not too far away from those thresholds will help in identifying their marginal value.³⁵ Note that the coefficients of R , *cumT*, and *Control* would not be identified in a static model.

5.4 Estimation

The estimation proceeds in several steps. First, I set the discount factor at 0.95. Second, I estimate the parameters that affect users' beliefs over the evolution of the state space, either in reduced form or nonparametrically. Third, I estimate the preference parameters.

³⁵Consider the example of section 5.3.2 with the threshold set at 2000 points and $\tilde{\alpha}$ such that $R_{t_0+3} - R_{t_0} = 200$. Then, line 3 of eq. 9 (i.e. the variation identifying β_6) is different from zero only for observations where $1800 \leq R_{t_0} \leq 1999$.

5.4.1 Estimation of beliefs' parameters

The estimation of the parameters that drive user beliefs exploits data at the answer level. The parameters $\{\gamma_1, \dots, \gamma_{13}\}$ drive beliefs on the expected number of edits, upvotes, and downvotes that an answer receives given its quality and the author's experience. Their estimation uses Poisson models on cross-sectional data, where each observation includes the number of edits, upvotes, and downvotes received by the answer on the publication week (dependent variables), the measure of quality of the answer, and the measures of experience of the author (see section 5.2). The probability π that a suggested edit is approved is the simple average approval rate. The rate of change in questions' availability is the OLS estimate of a trend. Finally, I estimate the decay parameters for the arrival of upvotes and downvotes on answers with a non-linear least squares estimation on panel data of the answer's histories in weeks from the publication date. Standard errors assume independence of each estimation.

5.4.2 Estimation of flow utility parameters without unobserved heterogeneity

For clarity, I first discuss estimation in the absence of unobserved heterogeneity. The estimation of the flow utility parameters proceeds with the following steps. First, I estimate the CCPs in reduced form. Because the state space is continuous, I cannot use a bin estimator to estimate the CCPs nonparametrically. Instead, I use a flexible L2-penalised multinomial logit model on scaled data between 0 and 1 and with saga solver.³⁶ Second, using the estimated CCPs and belief parameters, I compute the terms of the conditional value functions. More precisely, for each observation in the sample and each choice $\alpha \in \mathcal{A}$, I compute the terms of the sum in the differenced CVFs (see eq. 9, excluding the parameters). This step leads to the construction of a matrix with each column vector associated with one of the variables in the flow utility function. Third, I use such a matrix to estimate the flow utility parameters with a standard conditional logit model (McFadden (1974)).

³⁶As explanatory variables, I include R , R^2 , $cumT$, $Control$, the number of upvotes and downvotes expected to arrive from past contributions, the number of available questions, the experience (number of weeks spent on the site and number of answers posted), the period of participation and its squared value, the date (week) and its squared value, and the number of reputation points missing to reach the editing threshold (and its squared value). Appendix C.3 provides details.

5.4.3 Estimation of flow utility parameters with unobserved user types

I allow flow utility parameters to differ across two unobserved user types. For this purpose, I adapt the EM (Expectation Maximisation) algorithm proposed by [Arcidiacono and Miller \(2011\)](#) to my context. The algorithm maintains computational feasibility by searching for the values of the unobserved types and the flow utility parameters in sequential steps.

The algorithm proceeds with the following steps. First, I initialise the population probabilities for users to belong to an unobserved type, the differenced CVFs (computed as in section 5.4.2), and the flow utility parameters. To characterise unobserved user types, I vary the initial flow utility parameters such that one user type has higher costs of participation (i.e. lower marginal utility of C^A and C^E) than the other. Second, given the flow utility parameters, I compute the users' individual probability of being of a certain type. Third, with such individual probabilities, I update the population probabilities. Fourth, I estimate new CCPs for each user type, weighting the flexible multinomial logit with the individual probabilities. Fifth, I compute new differenced CVFs for each user type using the updated CCPs. Finally, I update the value of the flow utility parameters, maximising the following log-likelihood function:

$$\sum_{i=1}^N \sum_{\theta \in \Theta} \sum_{t=t_0}^{\mathcal{T}} \sum_{\alpha \in \mathcal{A}} q_{i\theta} \ln \left(\frac{\exp(\nu_{\alpha it}(z_{it}, \theta) - \nu_{\alpha^0 it}(z_{it}, \theta))}{\sum_{k \in \mathcal{A}} \exp(\nu_{k it}(z_{it}, \theta) - \nu_{\alpha^0 it}(z_{it}, \theta))} \right) \times d_{\alpha it} \quad (10)$$

where θ indexes user types and $q_{i\theta}$ is the probability that user i has type θ . With updated flow utility parameters, the algorithm restarts from step one and repeats until convergence.

5.4.4 Standard errors

I compute standard errors for the flow payoff parameters using a bootstrap procedure. In practice, given N users in the data, I sample with replacement N users and build 150 alternative datasets with their participation histories.³⁷ I then estimate the flow payoff parameters in each of these samples following the procedure described in section 5.4.3, and compute their standard deviation. This bootstrap procedure allows me to take into account the noise in both the reduced-form CCP estimation and the conditional logit estimation on the expected value terms. However, the standard errors do not incorporate the error in the belief parameters' estimates (used for the construction of the expected value terms) and, as a result, are possibly undervalued. The bootstrap procedure cannot incorporate the belief parameters' estimation, as it relies on different data.

³⁷In practice, the standard errors are computed with 146 estimations of the model.

5.5 Results

5.5.1 First-stage Estimates of Beliefs Parameters

Table 2 reports estimates of the parameters that drive users' expectations on the evolution of the state space, given their choices. The first set of estimates relates to the number of edits a user expects to receive from the community on an answer she publishes. Naturally, a higher-quality answer requires fewer edits. Estimates confirm this intuition, as higher quality and higher user experience lead to fewer edits. The second and third sets of estimates identify predictors of up-votes and down-votes. Again, estimates reflect what one would expect. Higher answer quality and higher user experience lead to more upvotes and fewer downvotes. Community edits instead increase the chances of both upvotes and downvotes. A possible justification is that community edits improve the answer, attracting more upvotes, but, at the same time, are more likely to occur on low-quality answers, which attract downvotes.

The parameter π reports the rate at which suggested edits get approved and shows that more than 80% of suggested edits get implemented. The parameters τ_U and τ_D characterise the exponential decay rate at which up-votes and down-votes continue to arrive in the following weeks after the publication of the answer. The small value of these parameters suggests a very steep decrease in votes with time. Finally, the rate of question availability indicates that the availability of questions has substantially increased over time.

5.5.2 Flow Payoff Parameters

Table 3 reports the estimates of the flow payoff parameters for each unobserved user type. Overall, all users have a positive marginal utility from accumulating reputation points and experience a direct cost (even though marginally decreasing) from contributing to answering and editing. Crucially, users gain value in contributing both answers and edits after they achieve control on editing, particularly for edits, which is consistent with the reduced form evidence. The estimate of the marginal utility from reaching the privileges ($cumT$) is negative, suggesting that, on average across privileges, users are not sensitive to the approach of the thresholds.

The estimation identifies two unobserved types: a smaller group of 478 users (about 19%), whom I call *committed*, and a larger group of 2147 users who I call *enthusiasts*. User types differ mostly in how they value participating in their contribution histories. *Committed* users have higher initial costs of editing than *enthusiasts*, but such costs decrease faster with more contributions, and are highly offset by extra marginal utility of editing after they gain control, which doubles that of *enthusiasts*. Preferences for answering are more similar across user types.

	Estimate	Std. Error
$\lambda_{E,j,t} = \exp(\gamma_0 + \gamma_1 QA_t + \gamma_2 Seniority_t + \gamma_3 Practice_t)$		
γ_0	-1.7871	0.04
γ_1	-0.1351	0.0066
γ_2	-0.0005	3e-05
γ_3	-0.0002	3e-05
$\lambda_{U,j,t} = \exp(\gamma_4 + \gamma_5 QA_t + \gamma_6 \lambda_{E,j,t} + \gamma_7 Seniority_t + \gamma_8 Practice_t)$		
γ_4	-0.5711	0.0083
γ_5	0.184	0.0012
γ_6	0.665	0.0046
γ_7	0.0002	3e-06
γ_8	-4e-05	3e-06
$\lambda_{D,j,t} = \exp(\gamma_9 + \gamma_{10} QA_t + \gamma_{11} \lambda_{E,j,t} + \gamma_{12} Seniority_t + \gamma_{13} Practice_t)$		
γ_9	-1.0762	0.0263
γ_{10}	-0.1528	0.0043
γ_{11}	0.5786	0.0187
γ_{12}	2e-05	1e-05
γ_{13}	-0.0003	1e-05
$ApprovedEdits_t \sim \mathcal{B}(NE_t, \pi)$		
π	0.8115	0.0033
$\Lambda_{U,t} = \Lambda_{U,t-1} e^{-\frac{1}{\tau_U}} + NA_t \lambda_{U,j,t}(QA_t)$		
τ_U	0.2297	0.0067
$\Lambda_{D,t} = \Lambda_{D,t-1} e^{-\frac{1}{\tau_D}} + NA_t \lambda_{D,j,t}(QA_t)$		
τ_D	0.2463	0.0092
$avail_t = \nu_0 + \nu_1 t + \epsilon_t$		
ν_1	108.6329	0.5489

Estimates of parameters governing belief formation processes over the state space's evolution. The $\{\gamma\}$ parameters are estimated with Generalised Least Squares Poisson models and capture, respectively, determinants of the number of edits on own answers, the number of up-votes, and the number of down-votes. π is estimated non-parametrically and captures the probability a suggested edit gets approved (on the whole sample of suggested edits and not only for users in the estimating sample). τ_U and τ_D are the decay times of up-votes and down-votes arriving on answers since answers' publication and are estimated via non-linear least squares. ν_1 is the rate at which the availability of questions grows over time and is estimated via ordinary least squares.

Table 2: First-Stage Estimates

However, *committed* users' marginal utility from answering is substantially higher once they gain autonomy on editing.

5.5.3 Unobserved user types characteristics

Table 4 reports descriptive statistics by user type. On average, *committed* users have consistently more information on their user profiles than *enthusiasts*. In particular, they have longer biographical descriptions, and they are more likely to display a full name (of the format *Name Surname*), a website, their location, and their LinkedIn profile. This suggests that *committed* users are possibly more extrinsically motivated, as more information makes them more recognisable both within and outside the online community.

Conditionally on editing, *committed* users provide lower quality edits, as measured by a large language model (LLM)'s assessment.³⁸ However, the rate at which their edits are rolled back (i.e. reverted to a previous version) and the rate at which their suggested edits are approved are not statistically different. This mixed evidence does not suggest a big difference in editing ability and effort across user types. Similarly, a reduced-form analysis of the effect of autonomy on editing quality shows no statistically significant differences across types, as shown in table 9 in appendix B.3.

5.6 Model Fit

To test the fit of the model, I forward simulate the data exploiting the same CCP-based representation of the value function used in estimation. For the exact same users in the sample and their exogenous states, the procedure sequentially computes the structural probabilities of making given contribution choices. For each period, it computes the structural probabilities, draws contribution choices from such probabilities, and updates the endogenous states of the following period. The structural probabilities correspond to the likelihood function (i.e. the expression within the round brackets in equation 10) evaluated with the type-specific flow-utility parameters for each choice $\alpha \in \mathcal{A}$. Such computation exploits the expression of the likelihood function in terms of the differenced conditional value functions (equation 8). This approach allows for the simulation of the data without making additional assumptions on the state space, and without having to set a terminal period.

³⁸The measure of edit quality captures the number of issues in the posts that have been fixed with the edit. I use Mistral AI. For each edit, I propose to the algorithm the versions of the post before and after the edit, and ask it to assess how many issues each has. I then compute the difference. Appendix A.2 provides additional details.

	Type: Committed	Type: Enthusiast	T-test Difference
R	0.0022 (0.0005)	0.0031 (0.0004)	-1.49
C^A	-0.2976 (0.0507)	-0.3091 (0.0547)	0.16
C^E	-2.9029 (0.2318)	-2.0027 (0.4069)	-1.94
$Tcum$	-0.1123 (0.0458)	-0.0779 (0.0237)	-0.72
$Control$	-0.1615 (0.8982)	-0.1439 (0.5949)	-0.02
$R \times Control$	0.0007 (0.0005)	0.0009 (0.0004)	-0.40
$C^A \times Control$	0.2331 (0.121)	0.2355 (0.0886)	-0.02
$C^E \times Control$	4.2768 (0.5368)	2.0908 (0.5196)	3.25
$(C^A)^2$	0.007 (0.0013)	0.0059 (0.0007)	0.69
$(C^E)^2$	0.2883 (0.0247)	0.2116 (0.0408)	1.74
$(C^A \times Control)^2$	0.0337 (0.0138)	-0.0015 (0.0021)	2.54
$(C^E \times Control)^2$	-0.4076 (0.0555)	-0.1879 (0.0544)	-3.13
Pop. probability of type	0.1883 (0.0667)	0.8117 (0.0667)	

Estimates are reported separately for each unobserved type. The column *T-test difference* reports the value of the t-statistic for the null hypothesis that the marginal utility of each variable is the same across user types, i.e. $H_0 : \beta_{X,\theta^C} = \beta_{X,\theta^E}$, for each variable X. Bootstrapped standard errors are in parentheses.

Table 3: Flow utility estimates

While the model approximates contribution patterns quite well, it under-predicts the total number of reputation points that users gain. Two main factors play a role in this misprediction. First, due to the discretisation of the choice set, the model is not able to replicate the highly skewed user contribution patterns.³⁹ It follows that a few users obtain many more reputation points than the model predicts. Second, users may receive points from sources not incorporated in the model, in particular from votes on questions. The unobserved types partly address the first issue, allowing for capturing more active contributors. To address the second issue, at each period in the simulation, I increase user accumulated reputation points with the points they received from their own questions.⁴⁰

Table 5 reports the model fit of the descriptive statistics, with user participation histories capped at 150 weeks (which is the length of the simulated histories). Besides the intervention mentioned above, the model under-predicts the total number of reputation points reached by some users and, as a consequence, the share of users reaching the threshold for control on editing. However, the qualitative

³⁹As a reminder, the largest number of weekly answers allowed in the model is 7. Whenever, in the data, users contribute between 4 and 122 answers, the model assumes a choice of 7 answers, which is the observed median value within that range.

⁴⁰This operation makes the simplifying assumption that users ask questions to receive answers, and not strategically to accumulate reputation points. As a consequence, they do not anticipate points arriving from questions.

	Type	mean	std. err.	median	n. users
Number of words in bio	Committed	30.481	2.852	4.000	478
	Enthusiasts	23.396	1.097	3.000	2147
Share of users with full name	Committed	0.272	0.020	0.000	478
	Enthusiasts	0.258	0.009	0.000	2147
Share of users with website	Committed	0.366	0.022	0.000	478
	Enthusiasts	0.350	0.010	0.000	2147
Share of users with location	Committed	0.559	0.023	1.000	478
	Enthusiasts	0.517	0.011	1.000	2147
Share of users with linkedIn	Committed	0.019	0.006	0.000	478
	Enthusiasts	0.012	0.002	0.000	2147
Share of users with term <i>english</i> in bio	Committed	0.111	0.014	0.000	478
	Enthusiasts	0.106	0.007	0.000	2147
Year of registration	Committed	2016.349	0.115	2016.000	478
	Enthusiasts	2016.128	0.049	2016.000	2147
Share of edits rolled back	Committed	0.008	0.003	0.000	312
	Enthusiasts	0.009	0.002	0.000	817
Share of edits rolled back (min. 3)	Committed	0.008	0.002	0.000	205
	Enthusiasts	0.009	0.002	0.000	415
Edit quality with LLM	Committed	1.081	0.035	1.039	312
	Enthusiasts	1.189	0.032	1.000	817
Edit quality with LLM (min. 3)	Committed	1.132	0.027	1.143	205
	Enthusiasts	1.212	0.024	1.200	415
Approval rate of suggested edits	Committed	0.853	0.013	0.950	307
	Enthusiasts	0.844	0.009	1.000	893
Approval rate of suggested edits (min. 3)	Committed	0.843	0.012	0.890	206
	Enthusiasts	0.851	0.009	0.923	433

Descriptive statistics on observable characteristics by user type. The first set of variables (up to the registration year) relates to information users disclosed in their profile pages at the time of the data download. The rest of the measures are proxies for user-level edit quality, including the rate at which users saw their edit reverted (*rolled back*), the rate at which users saw their suggested edits being approved, and a measure of edit quality computed with a large language model. Statistics marked with (*min 3*) are computed excluding users with less than 3 (suggested) edits.

Table 4: Characteristics of unobserved types

	sample	mean	std	min	median	max
Amount of Reputation Points Reached	data model	1295.15 410.27	4562.84 970.24	0.00 0.00	341.00 244.00	110252.00 19434.00
Average Answer Quality	data model	6.09 6.09	0.90 0.69	4.68 4.68	6.22 6.11	7.76 7.76
Number of Answers	data model	25.95 14.25	76.57 14.06	0.00 0.00	8.00 12.00	1038.00 210.00
Number of Edits	data model	10.76 3.15	59.88 9.65	0.00 0.00	0.00 1.00	1105.00 171.00
Reached Editing Privilege	data model	0.14 0.03	0.35 0.16	0.00 0.00	0.00 0.00	1.00 1.00

Sample size: 2625 users. Statistics on participation histories are capped at 150 weeks, with censoring at the download date as in the data. The *model* corresponds to the forward simulated data with two unobserved types.

Table 5: Model fit of descriptive statistics

insights are preserved. The simulated data reproduces well the reduced form results of section 4. Figure 11 in appendix C.4 replicates figures 2 and 3 and shows similar patterns.

6 Counterfactual Delegation Designs

As the allocation of control over the editing task affects users’ participation preferences, the quality and quantity of contributions depend on the delegation design. In this section, I simulate counterfactual contribution histories under alternative delegation thresholds.

The simulation approach is the same as the one exploited for the fit of the model, but varies the value of the editing threshold.⁴¹ In short, I forward simulate choices by, iteratively, 1) constructing the structural choice probabilities of each effort level, which exploit the reduced-form CCPs and the finite dependence paths, 2) drawing choices, and 3) updating the states. I simulate counterfactual participation histories for the same users and state values as in the data, but for different values of the reputation threshold, i.e. 0, 500, 1000, and 2000 points. The different counterfactual thresholds affect the number of points users need to reach the threshold and whether the user has autonomy. The other state variables are not impacted, if not through the counterfactual choices.

The adopted simulation approach allows substantial flexibility. Compared to

⁴¹For details, see section 5.6

backward recursion methods, the state space does not have to be constrained, the arbitrary length of the simulated history of participation does not affect individual choices, and the reduced computational burden allows for simulations of longer participation histories. However, it requires the reduced-form CCPs in the counterfactual environments, which are not observed. I predicted such CCPs using the model estimated with the observed data, assuming that the inclusion of the number of missing reputation points to the threshold as an explanatory variable allows me to recover CCPs sufficiently close to the counterfactual CCPs. The following section discusses in more detail the validity of this assumption.

6.1 Validity of the Counterfactuals

Let the reduced-form CCP estimator be defined as the function $p_{\alpha t}(x_t(\bar{R}), w_t; \beta)$. x_t captures the vector of state variables that depend on the value of the threshold for autonomy $\bar{R}$, namely the *Control* dummy ($\mathbf{1}\{R_{it} \geq \bar{R}\}$) and the number of points remaining to reach the threshold ($R_{it} - \bar{R}$), and w_t captures the remaining state variables. Such a function provides the probability that the user makes choice α in week t , and is estimated in reduced form with a multinomial logit model and the observed data (see appendix C.3).⁴²

Consider a counterfactual design where the reputation threshold to obtain autonomy is set at $\tilde{R} \neq \bar{R}$ points. To compute the structural choice probabilities of user choices, I use the reduced-form CCPs predicted as $\hat{p}_{\alpha t}(x_t(\tilde{R}), w_t; \hat{\beta})$. As a consequence, the validity of the counterfactuals rests on the key assumption that such predicted probabilities are valid, even though the parameters $\hat{\beta}$ were estimated using the original data, where the threshold is $\bar{R} \in \{1000, 2000\}$. In other words, the estimated reduced-form choice probability model has to predict the correct choice probabilities under a counterfactual threshold.

This condition is not violated under the following assumptions:

Assumption 1:

The threshold number of points affects users' decisions only by affecting the probability that users reach such a threshold.

In other words, a design with a different threshold affects user decisions only by changing the number of points the user needs to accumulate to obtain autonomy. This assumption implies that if, in a period t , the observed states have the same values in the original and counterfactual environments, the user choice probabilities should also be the same. This leads to the second assumption:

⁴²The change in threshold observed in the data allows us to separately identify both the coefficients of the reputation points (R_{it}) and that of the reputation points missing from the threshold ($R_{it} - \bar{R}$).

Assumption 2:

There does not exist an unobserved state that depends on the value of the threshold, and that affects user choices.

These assumptions would be violated if the counterfactual delegation design affects users' decisions in a different way than the observed one, and controlling for the observed states does not suffice to capture such differences. For instance, that would be the case if users perceive lower thresholds as less worth reaching, as they do not allow them to signal ability as much. Another relevant possibility is that the value of autonomy depends on how many users have it. Indeed, power may be more valuable if it is scarce. In that case, a lower threshold would make autonomy more accessible and less valuable.⁴³

Note that the estimation of the flow parameters already used a weaker version of these assumptions. Indeed, the autonomy threshold changed across time, and I assumed that the marginal utility of obtaining autonomy was the same for the two observed levels.

6.2 Counterfactual Results

Figure 4 reports the simulated average contribution histories of the users in the sample for 150 weeks of participation and by user type.⁴⁴

The simulations highlight the importance of the delegation design. Users have a general preference to contribute when they have control over editing. Autonomy in moderation increases the utility of participation on both editing and answering. As a consequence, users contribute more with lower delegation thresholds. Strictly positive delegation thresholds reduce the probability that users obtain autonomy on editing, leading to lower users' average contributions and a higher likelihood of no participation.

These patterns are, however, very different across user types. The majority of users (*enthusiasts*) are particularly active at the beginning of their contribution histories, suggesting an initial excitement or temporary motivation. As such, they reach the threshold faster and contribute more and higher-quality answers at the beginning. However, their contribution rate reduces substantially with time. With full delegation, more than 50% of them do not participate at all after almost three years of participation, and this number grows to 90% with delegation at 2000

⁴³However, I show that this is not the case. I look at users who did not lose the editing privilege when the website changed the threshold, and show that the sudden drop in the number of users with the privilege did not affect their contribution patterns. This result is reported in figure 12 in appendix C.5

⁴⁴The values reported are the weekly average of 10 separate simulations for the 2216 users in the sample who were observed for at least 150 weeks.

points. Note that this behaviour is not driven by anticipation of exit, but by a decreasing utility of participation across time. On the contrary, *committed* users (about 19% of the sample) contribute fewer answers at the beginning, but maintain more sustained contribution levels across time. In addition, their rate of inactivity is almost constant across time.

A major difference across types is also in the propensity for editing. *Committed* users are more likely to contribute edits across their participation history. As this propensity is much higher when users have autonomy on editing, the platform gives up on a substantial amount of community moderation by tying delegation of autonomy to performance.

Overall, delegation based on performance allows the platform to change the relative contribution shares of the different user types. More demanding performance thresholds decrease the average contribution of all users, but disproportionately so for *committed* users, who take a longer time to reach the threshold. This relative effect is, anyway, partly compensated by higher inactivity of *enthusiasts*. Table 6 illustrates this insight by reporting how much users of each type contribute over the simulated 150 weeks. Conditional on full autonomy over editing (i.e. 0-point delegation threshold), *committed* users contribute about 30% more answers than *enthusiasts*.⁴⁵ However, with a strictly positive threshold, the relation reverses and *enthusiasts* contribute 5-8% more answers than *committed* users. Users' average answer quality follows a similar pattern.

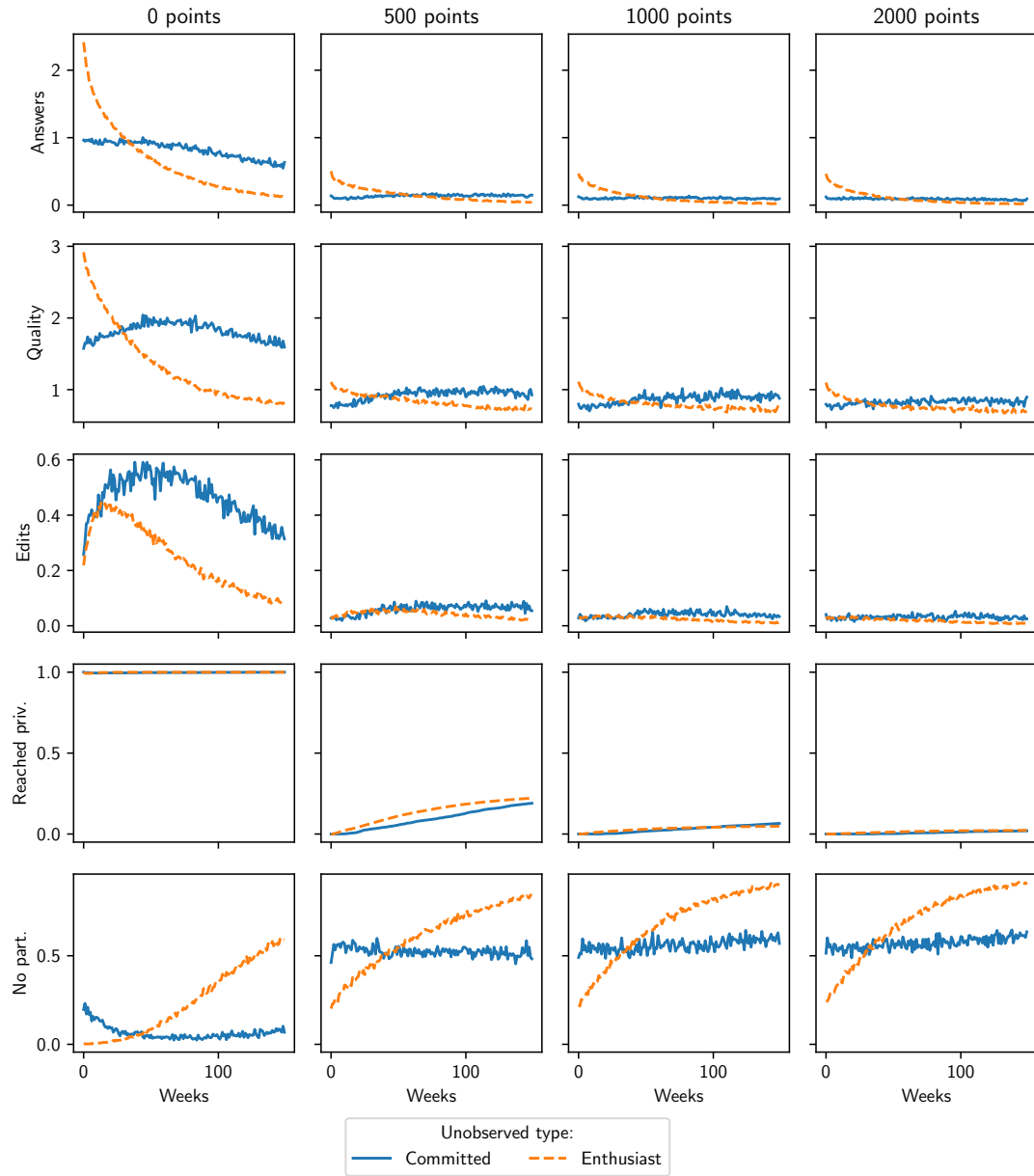
In terms of edits, the relative effect of delegation on performance on user types is different, as the value of autonomy on editing is much higher for *committed* users. Indeed, they always contribute more edits than *enthusiasts*. While the gap in edits halves (in percentage terms) when shifting the threshold from 0 to 500, more demanding thresholds increase the gap again. This is because with higher thresholds, the rate of inactivity of *enthusiasts* users increases much more than for *committed* users. With a threshold set at 500, *enthusiasts*' rate of achieving autonomy is 15% higher than that of *committed* users, but this gap disappears with a 2000-point threshold.

⁴⁵Note that in absolute terms, *enthusiasts* users contribute more as they compose the majority of the community

	Type	0 points	500 points	1000 points	2000 points
Answers	Committed	122.93	20.55	15.42	13.56
	Enthusiast	92.56	21.87	16.38	14.62
Quality	Committed	1.77	0.83	0.79	0.77
	Enthusiast	1.46	0.84	0.82	0.80
Edits	Committed	69.90	9.05	6.07	4.72
	Enthusiast	36.94	6.28	3.55	2.78
Reached priv.	Committed	1.00	0.19	0.07	0.02
	Enthusiast	1.00	0.22	0.05	0.02
No part.	Committed	0.07	0.52	0.56	0.57
	Enthusiast	0.24	0.61	0.68	0.70

Average user-level contributions for a simulated participation of 150 weeks, by delegation threshold. Includes the total contribution in answering (*Answers*), average answer quality conditional on answering (*Quality*), total contribution in editing (*Edits*), whether they reached the editing privilege (*Reached priv.*), and the rate of inactivity (average number of weeks of no participation - *no part.*)

Table 6: Simulated total contributions of a user



Simulated average weekly contributions by user type for different delegation designs (columns). The editing threshold is set to either 0, 500, 1000, or 2000 reputation points.

Figure 4: Simulated contribution paths by delegation threshold

7 Conclusion

This paper studies the delegation of autonomy based on performance targets in the context of an online community. It investigates voluntary users' preferences for autonomy and the tradeoffs the platform faces when being more or less demanding on the performance target.

First, in line with assumptions made in the theoretical literature (Aghion and Tirole (1997)), it shows that users' willingness to contribute to a given task depends on the level of autonomy the user has over the task. The paper finds that users post significantly more edits if they are directly implemented and do not require third-party approval. To my knowledge, this is new evidence in digital platform environments, which contributes to the growing literature that studies the role of autonomy as an incentive (Liberti (2018), Bandiera et al. (2021), Kala (2024)). Second, it finds that allocating autonomy over a task has a positive externality on other tasks. Indeed, the paper finds evidence that the utility from answering increases when users gain autonomy over editing. This result suggests that delegating control is an effective way to incentivise tasks whose outcomes are not easily measurable, i.e. where performance-based incentives may be distortionary (Holmstrom and Milgrom 1991). On the contrary, consistently with the theory, a more asymmetric allocation of autonomy in favour of answering leads to larger shares of answers on total contributions. Counterfactual simulations show that, relative to a full-delegation design, delegating autonomy at 2000 reputation points increases the proportion of answers on total contributions by about 15 percentage points. However, the overall level of contribution remains lower than under full autonomy.⁴⁶

Third, the paper finds heterogeneity in the value of autonomy and participation more generally. The small group of *committed* users value editing with autonomy much more than *enthusiast* users do, and sustain more contributions across time under all delegation designs. Conversely, *enthusiast* users are more likely to become inactive, but when delegation is based on performance, reach the delegation threshold faster and experience autonomy from an earlier stage of their participation history. These differences in the timing of their participation allow the platform to dynamically select *committed* users with higher delegation thresholds. However, there are no clear differences in editing abilities across types that would favour such a strategy.

Finally, the counterfactual with full delegation predicts the highest levels of contribution in both tasks, suggesting that career concerns either are not the main motives behind participation, or the allocation of autonomy does not necessarily

⁴⁶This effect is amplified by the asymmetry in reputation-based incentives, since answers allow to produce more reputation points than edits. However, suggestive counterfactuals that eliminate this asymmetry point to a limited amplification effect. See appendix C.5.1 for details.

signal users’ ability ([Holmström 1999](#)).

These results have important implications for platform design and the creation of user-generated public goods. They suggest that if platforms want to maximise user-generated content, they should provide as much autonomy as possible to users. This may partly explain Wikipedia’s success over the years, and is in line with more recent policies of Stack Exchange, which, since February 2023, set lower performance thresholds for future sites. However, the platform may find it optimal to delegate editing autonomy based on performance if it wants to give more weight to certain types of users in the overall content production.

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Appendix A Details on Data Construction

A.1 Answer Quality

Since the paper models the quality of an answer as a choice of the user, the associated empirical measure aims to capture the amount of effort the user has chosen, rather than an ex-post evaluation of the answer quality such as the number of upvotes the answer receives.

To construct this measure, I proceed with the following steps. First, I identify several possible proxies that capture different dimension of the answer quality. The proxies are calculated for each answer and include:

- The number of days separating the question’s publication date and the answer’s publication date. Because higher effort implies a quicker response, I make this measure negative, so that higher quality means less days.
- Text characteristics: 1) the number of words in the answer; 2) the share of informative words in the answer, computed as the number of words not included in the *stopwords* list over the total number of words; 3) the number of links in the answer; and 4) the number of pictures included in the answer.⁴⁷
- LLM-based measures of clarity, completeness, and informativeness. More precisely, I ask Mistral AI (version *mistral-small-2503*) to allocate to the answers one category for each aspect of quality. The option given are, respectively, 1) *Unclear*, *Partially clear*, and *Clear*; 2) *Unrelated*, *Partial answer*, and *Complete answer*; and 3) *Scarce*, *Informative*, *Deepened*. I mark as zero the lowest category in each dimension, and 2 the highest. Figure 5 displays the exact prompt.

Table 7 reports descriptive statistics of the proxies.

Second, I standardise each proxy variable and apply a Principal Component Analysis (PCA) transformation. The standardisation ensures that none of the variables will mechanically dominate in the computed components, as they originally take heterogeneous ranges of values. Third, I extract the first component as the measure of quality. The first component explains 34.21% of the variation. While this number is relatively small, there is a substantial drop in explained variation moving to the following components, as shown in figure 6.

⁴⁷The are several established lists of *stopwords* and I use the one available in the Python package *Natural Language Processing Toolkit*. Words included are, for instance, personal pronouns, prepositions, and conjunctions.

You will be presented with questions and answers on the use of the English language, extracted from a crowdsourced Q&A website. You will need to assess the quality of the answer. You will output ONLY a json object containing the following information:

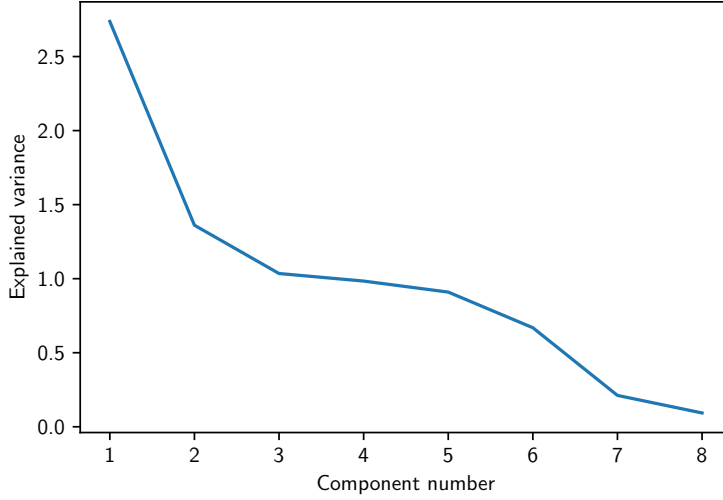
```
{
  clarity: string // degree of clarity of the answer.
  Choose only from the following list:
  1) Unclear; 2) Partially clear; 3) Clear.
  completeness: string // degree to which the answer addresses the question.
  Choose only from the following list:
  1) Unrelated; 2) Partial answer; 3) Complete answer.
  informativeness: string // degree of informativeness of the answer.
  Choose only from the following list: 1) Scarce; 2) Informative; 3) Deepened.
  explanation: string // justify your assessments with in a sentence.
}
```

Figure 5: Mistral AI classification prompt for answers’ quality

	mean	std	min	median	max
Days From Question (negative)	-48.06	238.35	-3297.00	0.00	0.00
LLM-based Clarity [0-2]	1.58	0.51	0.00	2.00	2.00
LLM-based Completeness [0-2]	1.55	0.51	0.00	2.00	2.00
LLM-based Informativeness [0-2]	1.44	0.62	0.00	2.00	2.00
Text-based Precision	0.49	0.07	0.00	0.48	1.00
Length	111.54	95.20	1.00	87.00	2855.00
Num. of Pictures	0.02	0.19	0.00	0.00	9.00
Num. of Links	0.34	0.92	0.00	0.00	82.00

Sample size: 147888 answers. The speed or response is the number of days between the question and the answer’s publication dates. It is made negative as less days suggest higher quality. The measure of *precision* is the number of informative words over the total number of words. The LLM measure were computed with Mistral AI.

Table 7: Proxy measures of answer quality



The variance explained by each component shows a substantial kink after the first component.

Figure 6: Explained variance of answer quality proxies by PCA components

A.2 Edit Quality (LLM)

The LLM-based measure of edit quality uses Mistral AI, version *mistral-small-2503*. The computation of the measure proceeds in the following way. For each edit, I recover the version of the post (question or answer) before and after the edit. I then prompt the LLM (Mistral AI) to assess how many issues the two versions have, where issues refer to grammatical errors, unclear writing, and lack of references and hyperlinks. The algorithm then outputs a number for each version. I take the difference between the number of issues found in the version before the edit and those found in the version after the edit, and I use this difference as a measure of edit quality. This procedure involves only edits on posts' bodies, as edits on tags and titles are too small and would lead to too noisy measurements. Figure 7 reports the system prompt used in querying the algorithm. It also asks for an explanation over the assessment. This follows results that found the LLMs more accurate if asked to justify their output.

A.3 Construction of scarcity variable

The scarcity variable is a measure of the quantity of available questions on the site at each point in time. It is defined as:

$$scarcity_t \equiv \frac{maxavail}{\log(avail_t)},$$

You will be presented with two versions of an answer[question]
extracted from a crowdsourced Q&A website.
Identify all the issues in both versions. Issues include grammar mistakes,
unclarity of the meaning, or lack of references and hyperlinks.
You will output ONLY a json object containing the following information:

```
{
version1: int // search and count the total number of issues in version 1.
version2: int // search and count the total number of issues in version 2.
explanation: string // explain which issues in a sentence without commas.
}
```

Figure 7: Mistral AI classification prompt for edit quality

where $avail_t$ is the number of unanswered questions on the website in week t , and $maxavail$ is the $\max\{\log(avail)\}$.

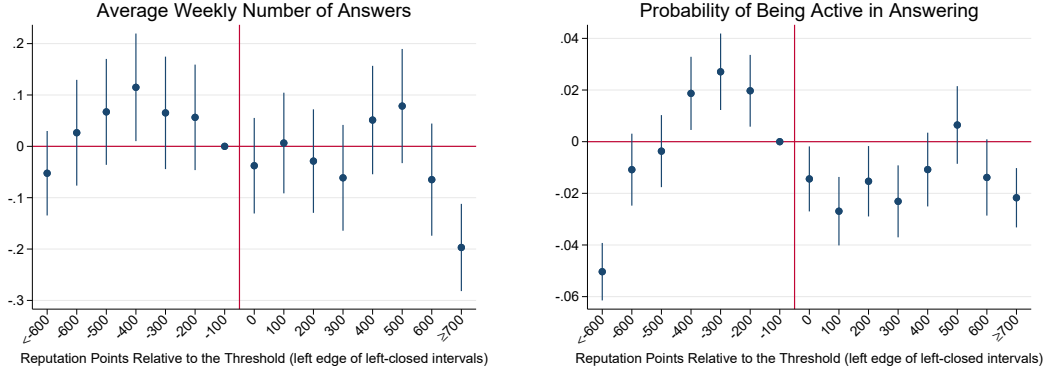
Appendix B Details on Reduced Form Results

B.1 The Effect of Autonomy on Edit Quantity

Table 8 reports the full list of estimates of the reduced form model discussed in section 4. In column 1, the dependent variable is the number of weekly edits while in column 2, it is a dummy equal to 1 if the user made at least one edit. These estimates are those reported in figure 2 in the paper. Columns 3 and 4 report estimates for models where the dependent variables are based on a different action (i.e. comments), which is not affected by the achievement of the privilege. Such dependent variables are, respectively, the number of comments made and a dummy equal to 1 if the user made at least one comment. Estimates show that editing increases after users achieve the privilege, while contributing via comments is not affected.

B.2 The Effect of Autonomy on Answer Quantity

Figure 8 reports estimates for the empirical model presented in section 4 with, as outcome variable, the number of answers produced (left in the figure) or a dummy equal to one if the user made any answer (right in the figure). These estimates suggest that users increase answering once the number of reputation points they have approaches the threshold. However, since users reach the threshold as a function of the number of answers they contribute, the timing of the discontinuity is endogenous in this case.



Reputation-points fixed effects before and after achieving the editing privilege. Sample of users who reached the threshold.

Figure 8: Answering Contributions Relative to Achieving the Privilege

	(1)	(2)	(3)	(4)
	NumEdits	At least one edit	NumComments	At least one comment
-700	-0.0470 (0.0635)	-0.0286*** (0.00423)	-0.449*** (0.0912)	-0.0742*** (0.00668)
-600	0.0151 (0.0795)	-0.00360 (0.00530)	0.00432 (0.114)	-0.0229** (0.00836)
-500	0.0164 (0.0796)	-0.000393 (0.00530)	0.129 (0.114)	-0.00787 (0.00837)
-400	0.0450 (0.0809)	0.00828 (0.00539)	0.149 (0.116)	0.0181* (0.00851)
-300	0.0844 (0.0845)	0.0150** (0.00563)	0.250* (0.121)	0.0331*** (0.00889)
-200	0.0802 (0.0793)	0.0210*** (0.00528)	0.117 (0.114)	0.0308*** (0.00834)
-100	0 (.)	0 (.)	0 (.)	0 (.)
0	0.0827 (0.0718)	0.0148** (0.00478)	0.0338 (0.103)	-0.0108 (0.00756)
100	0.132 (0.0757)	0.0202*** (0.00504)	-0.0378 (0.109)	-0.0281*** (0.00796)
200	0.113 (0.0778)	0.0213*** (0.00518)	0.0575 (0.112)	-0.00240 (0.00818)
300	0.156* (0.0796)	0.0256*** (0.00530)	0.0544 (0.114)	-0.00683 (0.00837)
400	0.194* (0.0815)	0.0294*** (0.00543)	-0.0543 (0.117)	-0.00802 (0.00857)
500	0.237** (0.0858)	0.0384*** (0.00572)	0.0429 (0.123)	0.00816 (0.00903)
600	0.216* (0.0843)	0.0249*** (0.00562)	0.00867 (0.121)	-0.00248 (0.00887)
700	0.398*** (0.0656)	0.0482*** (0.00437)	0.218* (0.0941)	0.00569 (0.00690)
NumAnswers	0.0688*** (0.00396)	0.0131*** (0.000264)	1.339*** (0.00448)	0.0439*** (0.000328)
NumComments	0.150*** (0.00183)	0.0125*** (0.000122)		
is_candidate	4.162*** (0.421)	0.304*** (0.0280)	3.057*** (0.605)	0.242*** (0.0443)
is_moderator	4.246*** (0.125)	-0.00252 (0.00835)	-3.032*** (0.180)	0.0330* (0.0132)
CopyEditor	30.68*** (0.639)	0.190*** (0.0425)	15.39*** (0.916)	0.223*** (0.0671)
StrunkWhite	7.756*** (0.311)	0.399*** (0.0207)	7.033*** (0.447)	0.213*** (0.0327)
Observations	146542	146542	146542	146542
User and Week FE	Yes	Yes	Yes	Yes

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 8: Estimates of reduced form model.

B.3 The Effect of Autonomy on Edit Quality

A reduced-form estimate of the impact of autonomy on edit quality suggests that users do not statistically change the quality of their edits once they reach the threshold. Column 1 of table 9 reports estimates of the following regression (excluding fixed effects):

$$EditQuality_{it} = \alpha_i + \alpha_t + \beta isEditor_{it} + \varepsilon_{it}$$

where α_i and α_t capture user and week fixed effects, and $isEditor_{it} \equiv \mathbf{1}\{r_{it} \geq \bar{R}\}$. The dependent variable is the average edit quality in weeks where the user made a positive number of edits, where the edit-quality measure is LLM-based (see appendix A.2 for details). The coefficient of interest β captures the average within-user change in edit quality after the user gains autonomy in editing. The sample excludes weeks with zero edits and users who did not reach the threshold. Column 2 studies heterogeneity across user types by interacting dummies for each user type.

The analysis shows that users' editing quality did not change significantly after they gained autonomy in editing. By replacing the variable $isEditor_{it}$ with reputation-points-ranges relative to the threshold fixed effects (i.e. $\{\beta_{r_{it}-\bar{R}}\}_{\forall r}$), estimates confirm this insight, as shown in figure 9.

	(1)	(2)
	edit_quality	edit_quality
isEditor	-0.0249 (0.0357)	
Committed x isEditor		0.0236 (0.0586)
Enthusiasts x isEditor		-0.0523 (0.0443)
Observations	9831	9831
User and Week FE	Yes	Yes

Standard errors in parentheses
 * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 9: The impact of autonomy on edit quality.



Reputation-points fixed effects before and after achieving the editing privilege.

Figure 9: The impact of autonomy on edit quality

Appendix C Details on the Structural model

C.1 Choice set

To reduce computational time for the estimation, I discretise the choice set to 21 possible combinations of effort, including a no participation decision.⁴⁸ All options in the choice set are listed in the table 10. The set of possible levels of contributions in answering is $\{1, 7\}$. A contribution of 1 answer corresponds to contributions between 1 and 3 answers in the data (which account for 70% of the weeks with positive contributions), while a choice of 7 corresponds to actual choices between 4 and 122 answers. 1 and 7 are the median values in the respective ranges. Similarly, for answer quality the possible choices are $\{4.68, 6.61, 7.76\}$, which correspond to the median values of the ranges $(0.517, 5.789]$, $(5.789, 7.386]$, and $(7.386, 10.819]$. The values 5.789 and 7.386 are the 33rd and 66th quantiles of the distribution of positive quality choices. Finally, users are allowed to contribute one or 9 edits, where a choice of one corresponds, in the data, to contributions between 1 and 4 edits (accounting for 75% of weeks with positive contributions), while the choice of 9 corresponds to contributions between 5 and 406 edits.

C.2 Details on Beliefs Assumptions

Section 5.2 describes functional form assumptions on the process of arrival of votes on answers. In particular, I assume that the arrival of up-votes and down-votes

⁴⁸A more natural assumption would be that users make discrete choices of task (answering and/or editing), and continuous choices for effort intensity. Bruneel-Zupanc (2020) provide a possible approach to estimate discrete-continuous choices.

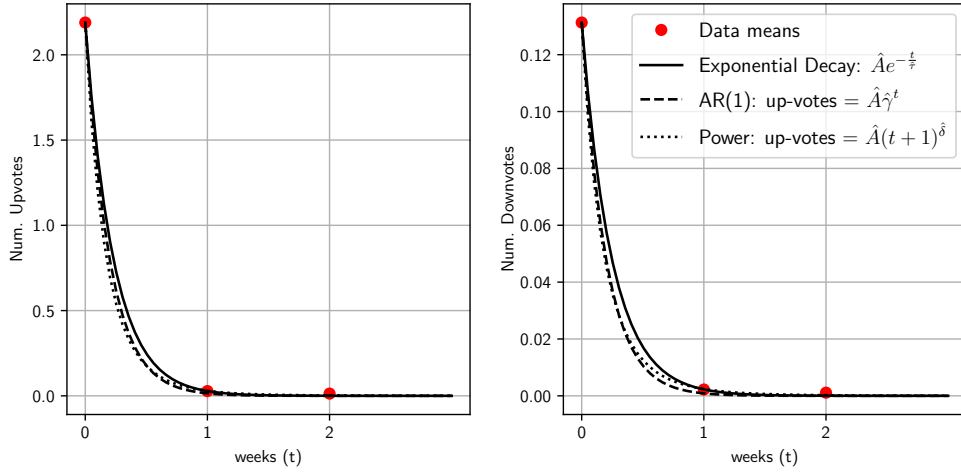
NA	0.00	0.00	0.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
QA	0.00	0.00	0.00	4.68	4.68	4.68	6.61	6.61	6.61	7.76	7.76
NE	0.00	1.00	9.00	0.00	1.00	9.00	0.00	1.00	9.00	0.00	1.00

NA	1.00	7.00	7.00	7.00	7.00	7.00	7.00	7.00	7.00	7.00	7.00
QA	7.76	4.68	4.68	4.68	6.61	6.61	6.61	7.76	7.76	7.76	7.76
NE	9.00	0.00	1.00	9.00	0.00	1.00	9.00	0.00	1.00	9.00	9.00

NA is the number of answers (in a week), QA is the average answer quality for the answers made, and NE is the number of edits (in a week). Values in the tables are rounded to the second decimal.

Table 10: Possible Combinations of Effort Levels

on an answer published on a given week follow an exponential decay process. This assumption reflects the fact that, on average, most up-votes and down-votes arrive on the same week the user publishes the answer, and very few votes arrive in the following weeks. Alternative functional forms that describe this sharp decrease in the number of votes across time would give similar predictions. Figure 10 shows how different alternative functional forms would fit the data.



Comparison of different possible functional form assumptions to describe the process of arrival of votes on answers, since publication week (i.e. $t = t_0$). Dots report data means across answers.

Figure 10: Arrival of Votes on Answers

C.3 Reduced Form Conditional Choice Probabilities

The reduced form conditional choice probabilities (CCP, $\hat{p}_{\alpha t}(z_t)$) are predicted probabilities that users make certain choices given certain states. To obtain such predicted probabilities I proceed in the following way:

1. Since the state variables z_t take values on very different ranges, I scale the data by subtracting their minimum and dividing by the difference between their maximum and their minimum. This leads to variables being all in the range $(0, 1)$, which eases the estimation.
2. I then estimate a multinomial logit model with response probabilities defined as:

$$p_{\alpha t}(z_t) = \frac{\exp(\beta_{0,\alpha} + z_t \beta_{\alpha})}{\sum_{k \in \mathcal{A}} \exp(\beta_{0,k} + z_t \beta_k)}$$

with:

$$\begin{aligned} z_t \beta_{\alpha} \equiv & \beta_0 R_{it-1} + \beta_1 \Lambda_{U,it-1} + \beta_2 \Lambda_{D,it-1} + \beta_3 \text{avail}_{it} + \beta_4 \text{AnswerNum}_{it} \\ & + \beta_5 \text{Seniority}_{it} + \beta_6 t + \beta_7 \text{date}_{it} + \beta_8 \text{cumT}_{it} + \beta_9 \text{Control}_{it} \\ & + \beta_{10} \Delta R_{it} + \beta_{11} R_{it-1}^2 + \beta_{12} t^2 + \beta_{13} \text{date}_{it}^2 + \beta_{14} \Delta R_{it}^2 \end{aligned}$$

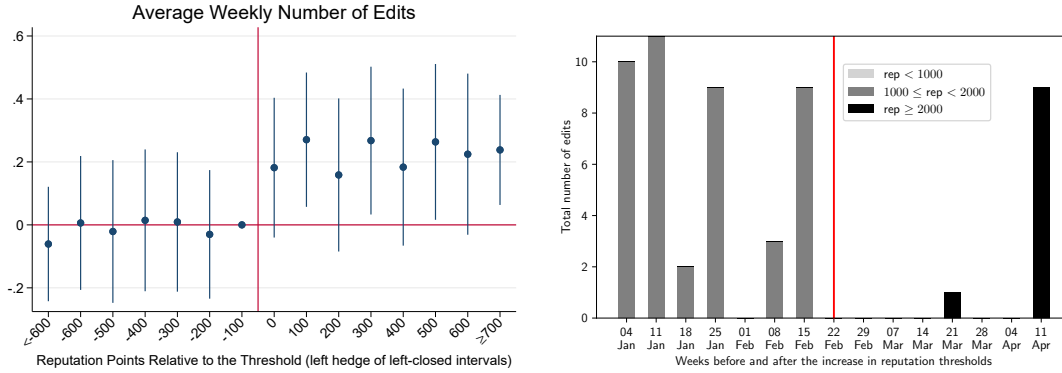
and where R is the number of reputation points, Λ_U and Λ_D are the expected number of up-votes and down-votes arriving from past effort, *avail* is the number of available questions to answer, *AnswerNum* is the number of answers already published up to period t , *Seniority* is the number of days passed since the registration day, *date* is the number of the calendar week since the first calendar week observed in the sample, *cumT* is the number of privileges obtained by the user, *Control* is a dummy equal to 1 if the user has autonomy and zero otherwise, and $\Delta R_t = R_t - \bar{R}$, where $\bar{R}$ is the reputation threshold for autonomy.

For the estimation I use the *LogisticRegression* function in the scikit-learn Python package, with l2 penalty term and *saga* solver.

3. I then predict the choice probabilities $\hat{p}_{\alpha t}(z'_t)$ for the possible future state realisations along the finite-dependence choice path.

C.4 Details on the Model Fit

Figure 11 reproduces the reduced form evidence of section 4 with the simulated data using the model.



The model correctly predicts the reduced form evidence. Users increase participation in editing once they gain autonomy, while the reduce it when they lose autonomy.

Figure 11: Replication of the reduced form evidence with simulated data

C.5 Details on Simulation of Counterfactuals

C.5.1 Relative contributions across tasks

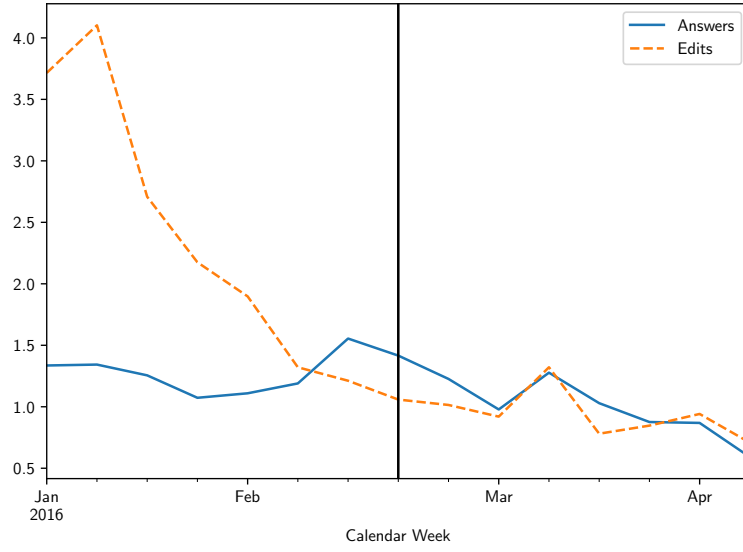
The delegation design affects the relative strength of incentives across tasks (editing and answering). When users do not have autonomy on editing, the incentives are particularly strong on answering, because users have autonomy in answering and receive more reputation points from answers. This asymmetry is partially reduced if and when users obtain autonomy in editing.

The simulation results reported in section 6.2 reflect these asymmetries, as users contribute relatively more answers than edits. Figure 13 reports the share of answers on total contributions of an average user, and the first four bars from the left refer to each counterfactual delegation threshold. It shows that more demanding performance thresholds to obtain autonomy on editing lead to relatively more answers than edits.⁴⁹

In an attempt to quantify how much of the higher propensity towards answering is due to asymmetry in autonomy-based incentives versus asymmetry in reputation-points-based incentives, I run an additional counterfactual that reduces the latter. In practice, I increase from 2 to 20 the number of reputation points that users obtain when one of their suggested edits gets approved.⁵⁰ Since an answer produces 21.6 reputation points on average in the first week after publication, the counterfactual simulates contributions in a context where users can receive a

⁴⁹Note, however, that in absolute terms, the full-delegation design leads to the largest amount of contributed answers.

⁵⁰Recall that approved edits give points if the user does not have autonomy in editing.

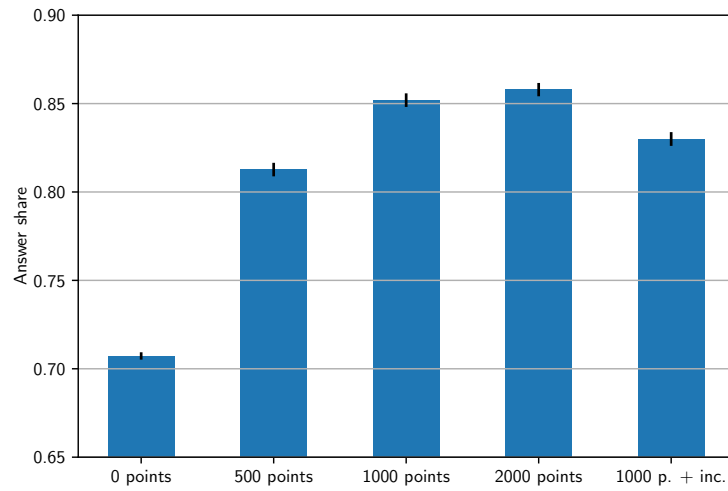


The graph reports the average number of contributions in answering and editing made by users who did not lose autonomy when the threshold changed. It shows that a sudden drop in the number of users with autonomy does not affect contribution patterns of users with autonomy.

Figure 12: Peer effect of autonomy

similar amount of points from both tasks, but have autonomy only on answering (until they reach the threshold). Note that the validity of this analysis requires strong assumptions and the results are only suggestive.

Simulation results are based on a 1000-points delegation threshold, and are reported in the right-most column of figure 13. They show that, by reducing the asymmetry in reputation-based incentives, the share of answers on total contributions decreases, but remains far above 50%, suggesting that the asymmetry in autonomy plays an important role in driving more answering than editing.



User-level share of answers on total contributions, by counterfactual delegation design. Bars report average values in each counterfactual design and 95% confidence intervals. The right-most bar refers to a counterfactual where 1) the delegation threshold is at 1000 points, and 2) users gain 20 points if one of their suggested edits gets approved, rather than 2. Users contribute substantially more answers than edits across counterfactuals. Increasing the reputation-based incentives on edits only partially offsets such asymmetry.

Figure 13: Share of Answers